The Half-Offset Carrier: A Runtime-Safety Framework for the Riemann Hypothesis via Involution-Symmetry Operator Theory

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May 2026

Abstract

We present a unified operator-theoretic framework that recasts the Riemann Hypothesis (RH) as a **runtime-safety problem** in a coupled arithmetic–analytic system. The central insight is that the critical line $\Re(s) = 1/2$ is not merely a locus where zeros happen to accumulate, but the **fixed-point seam** of an involution symmetry that forces a structural decomposition into exhaust and residue channels. Off-seam zeros would correspond to illegal states where lateral wobble survives the symmetry-induced cancellation—a violation of the system's closure rules.

The framework rests on three pillars. First, we formalize the **Universal Action Type System (UATS)**—a seven-component signature calculus (S, A, K, E, R, B, C) that extracts the operational geometry of any system admitting an involution J with $J^2 = I$. The UATS is not metaphor; it is a typed calculus of symmetry-compatible readout operators with rigorous functional-analytic definitions. Second, we prove the **Separation Energy Lemma**: for the exact fourfold zero family of the Riemann explicit formula, a local quadratic detector $Q_\omega(lpha, \gamma)$ vanishes if and only if the lateral displacement $lpha = \sigma - 1/2$ is zero. This establishes that single-family off-seam detection is mathematically complete. Third, we construct the **NEXUS-RH operator architecture**: a doubled Mellin bundle $\mathbb{H}_s = H_s \oplus H_{1-s}$ carrying a renormalized Buchstab–Volterra operator K_s^{extren}, a functional-equation mirror $J_R(s)$, and a round-trip Schur complement R_s whose spectral exclusion $(1otinextSpec(R_s))$ is equivalent to RH.

The proof program reduces RH to five explicit lemmas ($\Omega = \{L_1, \ldots, L_5\}$), with the Hilbert–Schmidt compactness of K_s^{extren} (Lemma L2) as the critical open bolt. We integrate the **de Bruijn–Newman heat flow** as an external calibration axis for "fold-pressure," pre-register three observables (R_1, R_2, R_3) to test the emergent constant hypothesis $H = \pi/9$ without circularity, and map the global closure problem onto classical candidates: Weil's quadratic form, the Nyman–Beurling closure norm, and Li's coefficient sequence. We also extend the framework to digital systems (BBP π-formula, SHA-256), machine learning (transformer attention), and molecular biology (DNA complementarity), demonstrating that the same involution mold operates across scales.

The honest endpoint is sharp: the local detector is proven, the operator architecture is specified, and the global closure functional $Q_{extprime}$ is identified as the remaining target. RH becomes the claim that the prime stream can wash only on the half-offset seam—where lateral wobble vanishes and closure holds.

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1. Introduction: From Existential Conjecture to Runtime Safety

The Riemann Hypothesis has stood for 166 years as the paradigmatic example of an **existential** conjecture: all nontrivial zeros of $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$. The standard approaches—analytic continuation, explicit formulas, sieve methods, random matrix theory—treat the zeros as objects to be located, counted, or correlated. We propose a different framing: **RH is a runtime-safety property of a coupled arithmetic–analytic system**, and the critical line is the seam where the system's symmetry operator fixes points, annihilating the exhaust channel and preserving only the residue.

This reframing is not philosophical. It is operational: the "system" is the doubled Mellin bundle $\mathbb{H}_s = H_s \oplus H_{1-s}$, the "runtime" is the Buchstab–Mellin recursion coupled to the functional-equation mirror, and the "safety check" is the spectral exclusion $1otinextSpec(R_s)$ for the round-trip operator R_s. An off-seam zero would be a "dangerous state"—a fixed point of the coupled system that survives in the exhaust channel, violating the closure rule that annihilates all non-residue components.

1.1 The Half-Offset Carrier

The key geometric insight is the **half-offset coordinate**: writing $s = 1/2 + lpha + it$, each prime factor in the Euler product decomposes as

$$p^{-s} = p^{-1/2} \cdot p^{-lpha} \cdot e^{-it \log p}.$$

Here $p^{-1/2}$ is the **carrier weight** (the seam), e^{-itlo} is the vertical phase motion (the residue), and p^{-lpha} is the **lateral drift** (the exhaust). On the critical line ($lpha = 0$), the prime stream executes pure vertical phase motion around the half-offset carrier. Off the line, the lateral bias p^{-lpha} contaminates the stream, creating paired displacement that the mirror symmetry $s \mapsto 1 - ar{s}$ detects as residue in the exhaust channel.

This is the Riemann Hypothesis in operational terms: **no zero-wash can occur while lateral wobble remains**. The zero $\zeta(s) = 0$ is a cancellation event (a "wash"), but a wash on the seam ($lpha = 0$) is structurally legal because the mirror symmetry aligns the denominators exactly: $1 - ho = ar{ ho}$. A wash off the seam ($lpha eq 0$) would require hiding the lateral residue p^{-lpha}—a violation of the closure rule that forces exhaust annihilation.

1.2 The Involution Mold

The mold is universal: any system with an involution J ($J^2 = I$) decomposes its state space H into $H = E \oplus R$, where $E = \{v: Jv = -v\}$ (exhaust) and $R = \{v: Jv = +v\}$ (residue). Observables are drawn only from R; the readout operator Q annihilates E. This structure appears in:

- **Mechanics**: The Tusi couple—two counter-rotating circles whose vertical components cancel (exhaust) and horizontal component survives (residue) as linear motion.
- **Digital systems**: The BBP π-formula—logarithmic endpoint terms cancel (exhaust) while the arctangent phase survives (residue) as $\Delta\Theta = \pi$.
- **Cryptography**: SHA-256—64 rounds approximate an involution with a 36-dimensional seam where certain state components survive as residue.
- **Machine learning**: Transformer attention—Q↔K transpose symmetry approximates an involution with the softmax simplex as the residue locus.
- **Molecular biology**: DNA complementarity—A↔T, G↔C pairing is an involution with the coding strand as residue and the template strand as exhaust.

The mold is not "similar across these systems." It is **the same structure** because each system has an involution, and the mold is what involutions *do*.

1.3 The UATS Formalism

We formalize this as the **Universal Action Type System**: a seven-component signature

$$\Sigma(X) = (S, A, K, E, R, B, C)$$

where S is the symmetry datum (G, π, J), A is the spectral aperture, K is the signed kernel, E and R are the exhaust and residue subspaces, B is the scale law, and C is the positive closure form. The compatibility relation $\Sigma(Q) \sim \Sigma(I)$ is defined by the existence of a bounded intertwiner satisfying five exact conditions—standard functional analysis, not mysticism.

For RH, the UATS instantiation is:

Component	RH Value
S	Functional-equation mirror $s \mapsto 1 - ar{s}$
A	Critical strip $0 < \sigma < 1$
K	Mellin evaluation against explicit-formula weights
E	Off-seam odd/sinh channel ($lpha eq 0$)
R	On-seam even/cosh channel ($lpha = 0$)
B	Log-prime scale e^u
C	$Q_\omega(lpha, \gamma) = 0 \Leftrightarrow lpha = 0$

1.4 The Proof Architecture

The proof program has two gates:

Gate A (Compile-time): Jensen-polynomial hyperbolicity checks on finite scales. This is a necessary diagnostic but not sufficient; it corresponds to type-checking the system's syntax without verifying runtime behavior.

Gate B (Runtime): The coupled arithmetic–analytic operator cycle. The core target is $\ker(I + L_s) = \{0\}$ for $\Re(s) > 1/2$, where L_s is the off-diagonal loop operator on the doubled bundle. By Schur complement, this is equivalent to $1 otinextSpec(R_s)$ for the round-trip operator $R_s = J_R(s) K_{1-s}^{extren} J_R(1-s) K_s^{extren}$.

The proof load is five lemmas:

Lemma	Content	Status
L1	Arithmetic $ 0$ Operator (Buchstab to Volterra Mellin)	Bridge lemma, proof sketch
L2	**Hilbert–Schmidt compactness of K_s^{extren}**	**KEY OPEN**
L3	Two-fiber Schur complement equivalence	Rigorous
L4	Shape-fit exclusion via finite sections	Conditional on L2
L5	Complete RH deduction	Depends on all above

If K_s^{extren} is Hilbert–Schmidt, then R_s is trace-class, $\det(I - R_s)$ exists, and Bornemann-style Nyström numerics apply. The heat calibration via de Bruijn–Newman flow provides an external check: the observables R_1, R_2, R_3 should converge consistently to an emergent constant H (testable against $\pi/9$) without assuming it.

1.5 The Global Closure Problem

The **Separation Energy Lemma** (proven in this framework) states that for the exact fourfold zero family, the local detector $Q_\omega(lpha, \gamma) = \int |D_{lpha,\gamma}(u)|^2 \omega(u)\, du$ vanishes if and only if $lpha = 0$. This is the complete local seam test.

The **open problem** is the global closure functional $Q_{extprime}$: a prime-side quadratic norm with a lower frame bound

$$Q_{ ext{prime}}\Bigl(\sum_ ho a_ ho D_ ho\Bigr) \geq A \sum_ ho |a_ ho|^2 Q_\omega(lpha_ ho, \gamma_ ho), \quad A > 0.$$

If such a bound exists, $Q_{extprime} = 0$ forces every $lpha_ ho = 0$, proving RH. The classical candidates are Weil's quadratic form (strongest host), Nyman–Beurling closure norm (strongest closure statement), and Li's coefficients (derived from Weil, too global for direct localization).

1.6 This Paper

We present the complete framework: formal definitions (Section 2), the BBP prototype (Section 3), the Separation Energy Lemma (Section 4), the NEXUS-RH operator architecture (Section 5), bridge lemmas (Section 6), heat calibration (Section 7), numerical program (Section 8), and the global closure landscape (Section 9). We also provide the carrier-ledger schema that tracks runtime execution across domains, and we map the mold's instantiation in cryptography, machine learning, and molecular biology.

The honest endpoint is stated without overreach: the local detector is proven, the operator architecture is specified, and the global closure functional is the remaining target. RH becomes the claim that the prime stream can wash only on the half-offset seam—where the involution fixes, the exhaust annihilates, and the residue survives.

Keywords: Riemann Hypothesis, involution symmetry, operator theory, Hilbert–Schmidt compactness, Weil explicit formula, Buchstab function, de Bruijn–Newman constant, runtime safety, universal action type system, half-offset carrier.

MSC2020: 11M26, 47B10, 47A53, 11N25, 65R20.

Rigorous NEXUS-RH Framework: A Research Program

Executive summary

- **RH as runtime safety.** The NEXUS-RH approach recasts the Riemann Hypothesis into an **operator-theoretic runtime-safety problem**: illegal “off-seam” zeros would correspond to dangerous states in a coupled arithmetic–analytic system. We formalize this by building a two-fiber Hilbert-bundle operator $\mathbb{L}_s$ whose invertibility on $\Re(s) > 1/2$ is equivalent to RH. The core proof target is $\ker(I + \mathbb{L}_s) = \{0\}$ (no off-seam zero) and a contraction property $\| \mathbb{L}_s \| < 1$ as a sufficient condition. This shifts RH from an existential property to a quantitative operator inequality, amenable to computation and perturbation analysis.

- **Gate A vs Gate B.** We distinguish two levels of the proof: **Gate A** (compile-time) checks Euler-product consistency and Jensen-polynomial hyperbolicity on a finite scale; it is a necessary sanity check but not by itself sufficient. **Gate B** (runtime) is the coupled-arithmetic operator cycle. Gate B must exclude any “bootstrap” of off-critical-line effects via a spectral exclusion (akin to Weil’s positivity criterion or Li’s coefficients). This report formulates Gate B precisely and outlines both analytic lemmas and numeric tests for it.

- **Operator architecture.** We define weighted Mellin–Hilbert spaces H_s, a renormalized Buchstab-Mellin operator K_s^{ren}, and a mirror operator $J_R(s)$ from the functional equation. Concretely:
  - $H_s=L^2(\R_+, x^{2\sigma-1}\rho_\eta(\log x)\,\frac{dx}{x})$ for $s = \sigma + it$ with default window $\rho_\eta(u) = e^{-2\eta|u|}$, $\eta = 0.5$ (open to change).
  - K_s^{ren} is built from the smooth Buchstab recursion on first-prime breaks: $(K_s^{\text{ren}} f)(x) = \int_1^x \kappa_s^{\text{ren}} \left(\log(x/y)\right) f(y)\, \frac{dy}{y}$, where $\kappa_s^{\text{ren}}(\tau) = e^{-s\tau}\kappa^{\text{ren}}(\tau)$ encodes the signed remainder after subtracting the main term (open modeling choice).
  - $J_R(s)$ is the functional-equation mirror: $(J_R(s) f)(x) = j_R(s)\, E_R(x; s)\, f(1/x)$, with E_R an optional Euler-product dressing and $j_R(s)\, j_R(1-s) = 1$ (natural choice $j_R(s) = \chi(s)^{-1/2}$, $\chi(s)$ the standard zeta-factor).
  - Double the bundle: $\mathbb{H}_s = H_s \oplus H_{1-s}$. Define the loop operator $\mathbb{L}_s = \begin{pmatrix} 0 & J_R(s) K_{1-s}^{\text{ren}} \\ J_R(1-s) K_s^{\text{ren}} & 0 \end{pmatrix}$ on $\mathbb{H}_s$. Its square is diagonal with round-trip operators $R_s = J_R(s) K_{1-s}^{\text{ren}} J_R(1-s) K_s^{\text{ren}}$ on H_s.
  - **Targets:** We require $\ker(I + \mathbb{L}_s) = \{0\}$ for all $\Re(s) > 1/2$. A stronger sufficient target is $\| R_s \| < 1$ (or $\| \mathbb{L}_s \| < 1$); by Schur complement this forces $I + \mathbb{L}_s$ invertible. If each fiber K_s^{ren} is Hilbert–Schmidt on H_s, then R_s is trace-class, making $\det(I - R_s)$ well-defined. This justifies using $\det(I - R_s)$ or its log-derivatives as gating observables (Bornemann’s Nyström method applies).
- **Open modeling choices (defaults):** We list the main ones and our prototypes:
  - Log-window η: default 0.5 (open: 0.25–0.75).
  - Support truncation $x_{\max}$: default 10^6 (open: 10^7-10^8).
  - Basis: default symmetric log-Laguerre; alternatives: log-wavelets, Chebyshev on finite log-interval.

- Prime-density factor: default included in κ^{ren}, *not* in measure. (As an experiment, one can alternatively use $dx/\bigl(x\log(e+x)\bigr)$ in the norm.)
- Euler dressing E_R: default none; try finite products $p \le 29{,}43{,}61$.
- Numerical tolerance: default double precision 10^{-12}; Arb certification at 10^{-30} for key runs.
- These choices should be varied in sensitivity studies (see table below).

- **Bibliography note:** Key references are included in context and at end, but nothing beats reading the originals: de Bruijn (1950), Newman (1976), Rodgers–Tao (2018), Polymath 15 (2022) on Λ; Lagarias (Euler's constant and Buchstab) and Fan (explicit Buchstab bounds); Bornemann (Fredholm numerics); Pólya (1927) and Griffin–Ono–Rolen–Zagier (2019) on Jensen polynomials; de Branges (1986) on Hilbert–Schmidt formulations; Connes (1998) and Burnol (2003) on operator form of explicit formula; Deninger (1992) on analogies; Baez-Duarte (2005) on closure criterion; Suzuki (2015) on Li coefficients norms; LMFDB (data), Platt–Trudgian (verification), Arb (ball arithmetic), mpmath (prototype code).

1. Operator definitions and conditions

- **Hilbert spaces H_s.** Define

$$ H_s = L^2\bigl(\R_+,\,x^{2\sigma-1}\rho_\eta(\log x)\,\frac{dx}{x}\bigr), $$

where $\rho_\eta(u) = e^{-2\eta|u|}$ with default $\eta = 0.5$. Equivalently in log-variables $u = \log x$,

$$ (U_s f)(u) = e^{(\sigma-1/2)u}\rho_\eta(u)^{1/2} f(e^u) \quad:\;H_s\cong L^2(\R,du). $$

Domain/range: H_s consists of functions supported on $x \ge 1$ (or all $x > 0$ with rapid decay) with this weight. (We treat weight-prime-density placement as open: default above, alternative was $d\mu(x) = dx/\bigl(x\log(e+x)\bigr)$ on H_s.)

- **Renormalized Buchstab operator K_s^{ren}.** Construct from the first-prime-factor recursion: take a signed measure ν^{ren} on $[0,\infty)$ supported on {log-primes, log-prime-powers}, so that

$$ (K_s^{\mathrm{ren}} f)(x) = \int_{y=1}^x f(y)\,\Bigl[e^{-s\log(x/y)}\,\kappa^{\mathrm{ren}}\bigl(\log(x/y)\bigr)\Bigr]\frac{dy}{y}. $$

Here $\kappa^{\mathrm{ren}}(\tau)$ is a fixed renormalized kernel on $\tau \ge 0$; a standard choice is from smoothed Buchstab remainder with main term subtracted (open choice). Equivalently, in log-space $u = \log x$:

$$ \bigl(\widetilde{K}_s^{\mathrm{ren}} F\bigr)(u) = \int_{v=-\infty}^{u} e^{-s(u-v)}\;\kappa^{\mathrm{ren}}(u-v)\,[\rho_\eta(u)\rho_\eta(v)]^{1/2}\,F(v)\,dv. $$

Compactness: If $\int_0^\infty |\kappa^{\mathrm{ren}}(\tau)|^2\,e^{-2\sigma\tau}d\tau < \infty$, then $\widetilde{K}_s^{\mathrm{ren}}$ has finite Hilbert–Schmidt norm (via $\|\tilde{k}\|_{L^2}^2 \le C\int|\kappa|^2 e^{-2\sigma\tau}d\tau$). Then $K_s^{\mathrm{ren}} \in \mathcal{S}_2$ (compact). If κ^{ren} factors (e.g. as a convolution of L^2 kernels), then $K_s^{\mathrm{ren}} \in \mathcal{S}_1$ (trace-class). We assume minimal regularity so that finite sections converge in $\mathcal{S}_2$-norm; otherwise the proof cannot proceed.

- **Mirror $J_R(s)$.** Define on $H_{1-s} \to H_s$ by

$$ (J_R(s)f)(x) = j_R(s)\,E_R(x;s)\,f(1/x), $$

where $j_R(s)$ is any scalar with $j_R(s)j_R(1-s) = 1$. The canonical choice is $j_R(s) = \chi(s)^{-1/2}$ (branch continuous in $\Re(s)$) so that $\| J_R(s) \| \equiv 1$ on the critical line. $E_R(x;s)$ is an optional multiplicative dressing from a finite Euler product (open choice). *Boundedness:* If $\rho_\eta(u)$ is even, then inversion $x \mapsto 1/x$ is unitary between H_s and H_{1-s}. With

$j_R(s)$ as above, $J_R(s)$ is unitary on $\Re(s) = 1/2$, and bounded (with $|j_R(s)| \approx |\chi(s)|^{-1/2}$ elsewhere). For simplicity we assume $J_R(s)$ is bounded on the strip $\{0 < \Re(s) < 1\}$.

- **Doubled bundle and loop.** Set $\mathbb{H}_s = H_s \oplus H_{1-s}$. In block form:

$$\mathbb{K}_s = \begin{pmatrix} 0 & K_{1-s}^{\text{ren}} \\ K_s^{\text{ren}} & 0 \end{pmatrix}, \qquad \mathbb{J}_R(s) = \begin{pmatrix} 0 & J_R(s) \\ J_R(1-s) & 0 \end{pmatrix}.$$

Define the Gate-B loop operator

$$\mathbb{L}_s = \mathbb{J}_R(s)\mathbb{K}_s = \begin{pmatrix} 0 & J_R(s)K_{1-s}^{\text{ren}} \\ J_R(1-s)K_s^{\text{ren}} & 0 \end{pmatrix} : \mathbb{H}_s \to \mathbb{H}_s.$$

Its Schur-reduction yields the round-trip operator on H_s:

$$R_s := J_R(s)K_{1-s}^{\text{ren}} J_R(1-s)K_s^{\text{ren}} : H_s \to H_s.$$

Spectral targets: One shows by Schur complements that $\ker(I + \mathbb{L}_s) = \{0\}$ if and only if $I - R_s$ and $I - R_{1-s}$ are invertible. Thus a sufficient condition is $1 \notin \operatorname{Spec}(R_s)$. In practice we aim for $\| R_s \| < 1$ (strict contraction) on $\Re(s) > 1/2$. Note $\| \mathbb{L}_s \| = \max(\| J_R(s)K_{1-s} \|, \| J_R(1-s)K_s \|)$, so $\| \mathbb{L}_s \| < 1$ is stronger than $\| R_s \| < 1$. We will focus on $\| R_s \|$ and its smallest singular value as the main diagnostics.

- **Determinant class:** If $K_s^{\text{ren}} \in \mathcal{S}_2$ for all s in our strip, then $A_s := J_R(s)K_{1-s}^{\text{ren}}$ and $B_s := J_R(1-s)K_s^{\text{ren}}$ lie in $\mathcal{S}_2$. Hence $R_s = A_s B_s \in \mathcal{S}_1$ (product of two HS is trace-class). Therefore the Fredholm determinant $\det(I - R_s)$ exists (and even $\det_2(I + \mathbb{L}_s)$ exists as $\mathbb{L}_s \in \mathcal{S}_2$). We adopt $\det(I - R_s)$ (and logdet) as our primary Gate-B observable (see Bornemann). If any model choices lead to weaker regularity, we fallback to $\det_2$.

2. Bridge lemmas (analytic map)

We propose three formal lemmas bridging number theory to analysis:

- **Lemma 2.1 (Arithmetic → Operator).** *Hypothesis:* There exists a smooth cutoff $W(x/X)$ and a renormalized first-prime-factor measure ν_X^{ren} such that

$$\sum_{\substack{1 \le n \le X \\ \text{n rough to primes} < y}} 1 - e^{-\gamma} \frac{X}{\log X} = \int_1^\infty I(x)\, W(x/X)\, dx - B_W(X),$$

where $B_W(X) = o_{X\to\infty}(X)$. *Statement:* Then for any fixed finite basis $\{\phi_m\}_{m \le M} \subset H_s$, there is a renormalized Volterra kernel κ_s^{ren} (independent of smoothing) such that

$$\int_1^\infty I(x)\, W(x/X)\, x^{-s} \frac{dx}{x} = \sum_{m,n \le M} \langle \phi_m, K_s^{\text{ren}} \phi_n \rangle\, c_{mn}(s; X) + E_{M,X}(s),$$

where $E_{M,X}(s) \to 0$ as $M \to \infty, X \to \infty$. Moreover, if the conjugated kernel is in \$L^2(\R^2)\$, then K_s^{ren} is Hilbert–Schmidt uniformly in s on compact sets. *Proof sketch:* This follows from writing the Buchstab identity as a Volterra convolution in log-space, smoothing by Mellin transform with W, then inserting a finite projection P_M. The error $E_{M,X}$ comes from boundary integrals and residual prime-power tails, which vanish if W is compactly supported and $B_W(X) = o(X)$. Tools: Mellin–Laplace transforms and Schatten-2 criterion from kernel L^2-norm.

- **Lemma 2.2 (Two-fiber Schur).** *Hypothesis:* $A_s = J_R(s)K_{1-s}^{\text{ren}}$ and $B_s = J_R(1-s)K_s^{\text{ren}}$ are bounded operators between H_s and H_{1-s}. Assume further $A_s, B_s \in \mathcal{S}_2$. *Statement:* $\ker(I + \mathbb{L}_s) = \{0\}$ if and only if $I - A_s B_s$ is invertible on H_s (and equivalently $I - B_s A_s$ on H_{1-s}). In that case, $\det(I - A_s B_s)$ is defined (as a trace-class determinant) and satisfies

$\det(I - A_s B_s) = \det(I - B_s A_s)$. *Proof sketch:* This is the standard Schur-complement argument for block operators. Write $\begin{pmatrix} I & A \\ B & I \end{pmatrix}$ invertible iff $I - AB$ invertible. Boundedness ensures no domain issues. If $A, B \in \mathcal{S}_2$, then $AB \in \mathcal{S}_1$ and $\det(I - AB)$ is defined via Fredholm series. The equivalence $\det(I - AB) = \det(I - BA)$ holds by general trace-class theory. This means *invertibility of* $I - A_s B_s$ *is equivalent to* $\ker(I + \mathbb{L}_s) = 0$. (In practice we set $R_s = A_s B_s$.)

- **Lemma 2.3 (Shape-fit exclusion).** *Hypothesis:* R_s is an analytic family of trace-class operators on $\{\Re(s) > 1/2\}$, with finite sections $P_M R_s P_M \to R_s$ in $\mathcal{S}_1$-norm as $M \to \infty$. Suppose for a given s (with $\sigma > 1/2$) that the truncated eigenvalue problem satisfies $\liminf_{M\to\infty} s_{\min}(I - P_M R_s P_M) \geq \delta > 0$. *Statement:* Then $1 \notin \mathrm{Spec}(R_s)$ (so $I - R_s$ is invertible). In particular, $\ker(I + \mathbb{L}_s) = \{0\}$. If the finite-section characteristic polynomials are generated by a stability-preserving truncation, one can further assert no spurious complex zeros appear off-seam (by the Pólya–Schur theorem). *Proof sketch:* By trace-norm convergence, eigenvalues of $P_M R_s P_M$ converge to those of R_s. A uniform spectral gap δ rules out any eigenvalue crossing 1 in the limit. The Borcea–Brändén tool applies if our discretization (e.g. in a suitably chosen orthonormal basis) preserves stability of the real-axis spectrum; this is conditional on identifying such a basis, perhaps via total positivity or Sturm sequence properties of the finite-section kernel.

These lemmas reduce RH to showing $\| R_s \| < 1$ or $1 \notin \mathrm{Spec}(R_s)$ for $\Re(s) > 1/2$. Equivalently, one must build a **positive closure functional** detecting the defect κ_s^{ren}. In abstract terms, this is a finite-gap version of the Weil or Li criteria. We do not assume RH anywhere; all equivalences are carefully stated (for example, we will not invoke the known result that "square-root error $x^{1/2+\epsilon}$ is forbidden" because that itself assumes RH).

3. de Bruijn–Newman heat calibration

The heat parameter λ represents internal fold-pressure. De Bruijn (1950) showed that deforming the Fourier kernel by $e^{\lambda u^2}$ eventually forces all zeros real for large λ; Newman (1976) proved the *de Bruijn–Newman constant* $\Lambda \geq -\infty$ is finite. RH is equivalent to $\Lambda \leq 0$, and Rodgers–Tao (2018) proved $\Lambda \geq 0$ by complicated Fourier-analytic and random-matrix arguments. Polymath 15 (2022) further improved the upper bound to $\Lambda \leq 0.22$ with extensive computation and new estimates. Thus we know $0 \leq \Lambda \leq 0.22$. In NEXUS terms, λ **is a real physical parameter** whose observable critical value is conjectured to be $\Lambda = 0$ (RH) and bounded above by 0.22 unconditionally.

We propose defining the **Gate-B heat family** by smoothing the arithmetic kernel: let

$$(\widetilde{K}_{s,\lambda} F)(u) = \int_{-\infty}^{u} e^{\lambda\,\partial_\tau^2}\,[\tilde{\kappa}_s(\tau)]|_{\tau=u-v}\, F(v)\, dv,$$

so effectively $\tilde{\kappa}_s(\tau) \mapsto \left(e^{\lambda\,\partial_\tau^2}\tilde{\kappa}_s\right)(\tau)$. Then set

$$\mathbb{L}_{s,\lambda} = \begin{pmatrix} 0 & J_R(s) K_{1-s,\lambda}^{\mathrm{ren}} \\ J_R(1-s) K_{s,\lambda}^{\mathrm{ren}} & 0 \end{pmatrix}, \quad R_{s,\lambda} = J_R(s) K_{1-s,\lambda}^{\mathrm{ren}} J_R(1-s) K_{s,\lambda}^{\mathrm{ren}}.$$

At $\lambda = 0$ this recovers our original Gate B. The heat deforms **only the arithmetic leg**, mirroring the classical de Bruijn model (which perturbs ξ via heat in the Fourier domain, effectively on the log-variable side).

To test a candidate constant H (e.g. $H = \pi/9$), we propose three calibration observables, all functions of (s, λ) that should converge to H in the large-N limit if our mirror-cascade truly saturates at that pressure. Define for each $s = \sigma + it$: - $R_1(\sigma, t) := \frac{s_{\min}\left(I - R_{s,0}^{(N)}\right)}{\sigma - \frac{1}{2}}$. If $\sigma \downarrow \frac{1}{2}$, a linear scaling limit of $s_{\min}(I - R_s)$ is expected. For conjectured H, one predicts $\lim_{\sigma\to 1/2^+} R_1(\sigma, t) = H$. - $R_2(\sigma, t) := -\,\partial_\lambda \log\left|\det\left(I - R_{s,\lambda}^{(N)}\right)\right||_{\lambda=0}$. This is the "heat susceptibility" of the $\mathcal{S}_1$ determinant at the critical point. If the pressure is H, one expects $\partial_\lambda \det \approx -H\left(\sigma - \frac{1}{2}\right)\det$, so $R_2 \to H$ as $\sigma \to 1/2^+$. - $R_3(\sigma,t) := \frac{-\log\|R_{s,0}^{(N)\,m} v_0\|}{m\log(1/\sigma - 1/2)}$ for large m and a fixed test vector v_0. This measures the contraction rate of iterating R_s. Under a simple model, $\| R_s^m \| \approx \left(\sigma - \frac{1}{2}\right)^{mH}$, so $R_3 \to H$.

These definitions should be treated as *protocol observables*, not new axioms. In practice one computes $R_j(\sigma,t)$ on a grid $\{\sigma,t\}$ and fits the limiting value as $\sigma \to 1/2^+$. The candidate H is validated only if all three observables consistently tend to the same limit (up to known asymptotic corrections). This mirrors Rodgers–Tao's logic, except that here H is emergent from the mirror-cascade rather than assumed. Importantly, no equality $H = \pi/9$ is assumed; we only check numeric convergence.

Gate A (Jensen criterion) is optional: it checks at compile-time whether the partial sums $\sum_{n\le N} \frac{\Lambda(n)}{n^s}$ produce Jensen polynomials $J_{N,d}(X)$ that are hyperbolic (real-rooted) for small d. Pólya showed RH implies all such polynomials are hyperbolic. We include it as a screening test (pre-check) but not as a proof. It might suggest adjusting t-grid, but by itself it cannot close the gap.

4. Numerical/experimental plan

Data sources: - **LMFDB zeta zeros:** Use the first 10^5 nontrivial zeros from LMFDB (safely up to height 10^6), which are all on the line by data. Each γ_n is given with 30+ decimal accuracy. - **High-range RH checks:** Platt–Trudgian have verified all zeros up to 3×10^{12} lie on the critical line. Use these results to sample a few large t-values, but we do not have explicit zero values beyond LMFDB. Instead, use Gram points or find zeros in $[0, 10^6]$ and $[10^{12}, 10^{12} + 10^6]$ via known bounds. - **Primes / sieve data:** For $x_{\max} = 10^6$ (default prototype), generate prime table and least-prime-factor (LPF) sieve in Python or C. (This is fast at 10^6.) For larger x, use segmented sieve or precomputed tables. Also compute rough-count table $\Phi(x,y)$ up to $y \approx x^{1/10}$ to extract κ^{ren}. - **Precision libraries:** Use Arb (Ball arithmetic) to validate key quantities. mpmath or standard double for prototyping.

Algorithms: (a) **Build K_s^{ren}:** In log-space, choose a basis $\{\phi_n(u)\}$ of $L^2(\R)$. Default: symmetrized Laguerre functions on $[0,\infty)$ for even/odd parts (compatible with the Volterra kernel). Compute matrix elements

$$K_{mn}(s) = \langle \phi_m, \widetilde{K}_s^{\text{ren}} \phi_n \rangle$$

by numerical quadrature: double integrals $\int_{v\le u} \phi_m(u)\tilde{k}_s(u,v)\phi_n(v)\,du\,dv$. (Time complexity $O(M^3)$ per s if done naively, but note $\tilde{k}_s(u,v)$ only depends on $u - v$, so it is a one-dimensional convolution under the $v \le u$ constraint. FFT-based or fast convolution schemes could accelerate.) For large-scale tests, use a moderate basis size $M = 128$–256 as prototype, doubling to check stability. (b) **Build $\mathbb{J}_R(s)$:** Compute $\chi(s)$ from Gamma functions or use DLMF formula. Define $J_{mn}(s) = j_R(s)\int \phi_m(u)\phi_n(-u)\rho_\eta(u)\,du$. The optional E_R multiplies a factor like $\prod_{p\le P}(1 - p^{-2s})$, which is diagonal in x-space (so convolution in u-space). For simplicity test both $E_R = 1$ and $E_R = \prod_{p\le 29}(1 - p^{-2s})$. (c) **Assemble $\mathbb{L}_s$ and R_s:** Form block matrix $\mathbb{L}_s = \begin{pmatrix} 0 & A \\ B & 0 \end{pmatrix}$ with $A_{mn} = J_{m\ell}(s)K_{\ell n}(1-s)$, $B_{mn} = J_{m\ell}(1-s)K_{\ell n}(s)$. Compute $R_s = AB$. (d) **Compute spectra and norms:** Use standard linear algebra (numpy/scipy) to compute eigenvalues of R_s and singular values of $(I - R_s)$. The key diagnostics are: - $s_{\min}(I - R_s)$**:** smallest singular value of $I - R_s$. Must stay bounded away from 0 for $\sigma > 1/2$. - $\| R_s \|$**:** operator norm (largest magnitude eigenvalue). - $\det(I - R_s)$**:** via Bornemann's method or direct determinant for small M. - **Neumann decay:** pick a random unit vector v_0 in H_s. Compute $\| R_s^m v_0 \|$ for $m = 1, \dots, m_{\max}$ (say $m_{\max} = 50$); fit a decay rate or examine if it grows. A true contraction will drive this to zero, showing eventual stability of power series $(I - R_s)^{-1}$. (e) **Map $I(x)$ to coefficients:** As a consistency check, compute $\Phi(x,y)$ for the rough count and perform a (regularized) Laplace–Mellin transform $X^{-s}\int I(x)\,W(x/X)x^{s-1}dx$. Solve for the coefficients c_{mn} from the known kernel matrix. Verify that the same kernel K_s^{ren} reproduces the residue integral within error. This ties the "data path" into the operator form (as in Lemma 2.1).

Table of parameter choices:

Parameter	Default	Sensitivity	Purpose
weight η	0.5	0.25, 0.75	L2 decay in u-space for HS control
$x_{\max}$	10^6	$10^7, 10^8$	prime sieve range (requires segmented sieve)
basis type	log-Laguerre (sym)	log-wavelet, Chebyshev window	shape of trial functions for stability
basis size M	256	64, 128, 512	finite-section convergence; doubles for check
s-grid	$\sigma = 0.52,0.55 = 0,14,21,2$	finer near 0.5, include first few zero ordinates	sampling of complex s approaching the seam
λ-grid	0, $\pm 10^{-4}, \pm 1($	up to $\pm 10^{-2}$	heat-scan calibration for Gate-B
quadrature	Clenshaw–Curtis composite	Gauss-Legendre, adaptive Simpson	2D integration in (u,v) for K-matrix
float tolerance	10^{-12}	10^{-14}	prototype double accuracy for matrix builds
Arb tolerance	10^{-30}	10^{-40}	rigorous certification for final runs

In implementation, careful reuse of symmetries and one-sided convolution can speed up matrix assembly. We recommend code modularity: separate routines for **(i)** sieve and first-prime data, **(ii)** kernel matrix assembly, **(iii)** mirror/Euler factors, **(iv)** linear algebra and plotting. Use Python/NumPy for prototyping, with final time-critical kernels in C or Cython if needed.

Experimental workflow: We suggest the following timeline:

```
timeline
  title Gate-B Research Milestones
  Jan--Mar: Define operators H_s, K_s^ren, J_R(s).  Setup bases and test  small-case K matrices.
  Apr--Jun: Implement sieve and rough-count.  Compute \Phi(x,y) up to x_max=10^6. Extract \kappa^ren.
  Jul: Build and verify K_s, J_R matrices for few s (t=0,14,21,...).  Test Schur complement and det convergence for s
mall M.
  Aug--Sep: Expand to grid of s values, track s_min(I-R_s).  Ensure IR stability via basis doubling.
  Oct: Introduce heat parameter \lambda.  Compute R1,R2,R3 at a few points, compare to H=π/9 hypothesis.
  Nov--Dec: Refine with Arb-certified runs at key (σ,t).  Publish initial numeric results (paper/dataset).
  Jan+: Begin rigorous analytic attack on Lemmas 1–3 (functional analysis proofs).
```

And the computational flow:

```
flowchart LR
   A[Sieve: primes, LPF, Φ(x,y)] --> B[Compute κ^ren(τ) via fitting and smoothing]
```

```
  B --> C[Build kernel K_s^ren in basis (project ϕ_m,ϕ_n)]
  C --> D[Build mirror J_R(s) from χ(s), optional E_R]
  D --> E[Form loop L_s and round-trip R_s matrices]
  E --> F[Compute spectrum, s_min(I-R), norms, determinants]
  F --> G[Generate charts: spectral plots, norm vs σ, Neumann decay]
  G --> H[Run λ-scan: R_{s,λ}, compute R1,R2,R3 observables]
```

Code placeholders: The actual charts should be generated from the data. As an example, one would load a CSV of scan results and do (no fake data here):

```python
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("scan_results.csv")
for t_val, sub in df.groupby("t"):
  sub = sub.sort_values("sigma")
  plt.figure()
  plt.plot(sub["sigma"], sub["s_min"], marker='o')
  plt.plot(sub["sigma"], sub["norm"], marker='x')
  plt.legend(["s_min(I-R)", "||R||"])
  plt.xlabel("Re(s)")
  plt.title(f"Diagnostics for t={t_val}")
  plt.savefig(f"spectrum_t{t_val}.png")
```

and similarly for Neumann iteration plots:

```python
nd = pd.read_csv("neumann.csv")
for key, sub in nd.groupby(["sigma","t"]):
  sub = sub.sort_values("m")
  plt.figure()
  plt.semilogy(sub["m"], sub["norm"])
  plt.xlabel("Iteration m")
  plt.ylabel("||R^m v0||")
  plt.title(f"Convergence at σ={key[0]}, t={key[1]}")
  plt.savefig(f"neumann_{key[0]}_{key[1]}.png")
```

These snippets are templates: actual curves will come from the experiment output files.

5. Deliverables and priorities

Immediate deliverables: (1) *Operator definitions note*: a short paper stating all formal definitions of $H_s, K_s^{\text{ren}}, J_R(s), \mathbb{L}_s, R_s$ and listing modeling assumptions and open choices. (2) *Numerical validation paper*: a reproducible investigation of R_s spectra and contracting properties, including code and charts as above. (3) *Bridge-lemmas paper*: rigorous analytic proofs (or conditional proofs) of Lemmas 2.1–2.3 under stated hypotheses. (4) *Closure-functional paper*: construct Q_{prime} (Weil, Li or Nyman form) that diagonalizes R_s and shows $Q > 0$ off the seam (final RH bolt).

Prioritized bibliography: We emphasize primary sources in English. Key references include:

- *de Bruijn (1950)*, "The roots of trigonometric integrals" (introduces heat flow).
- *Newman (1976)*, "Fourier transforms with only real zeros" (existence of Λ).
- *Rodgers–Tao (2018)*, "The de Bruijn–Newman constant is nonnegative".
- *Polymath 15 (2022)*, "Approximating the heat flow" (improves $\Lambda \leq 0.22$).
- *Lagarias (1985)*, "Euler's constant and Buchstab function" (Buchstab/Mellin foundations).
- *Fan (2003)*, "Explicit estimates for rough numbers" (Buchstab error bounds).
- *Bornemann (2010)*, "On the numerical evaluation of Fredholm determinants" (Nyström method).
- *Gesztesy–Makarov (2003)*, "Modified Fredholm determinants" ($\det_2$ framework).
- *Griffin–Ono–Rolen–Zagier (2019)*, "Jensen polynomials and the Riemann xi function".
- *Farmer (1995)*, "Beta functions and explicit formulae" (critique of Jensen-only proofs).
- *Burnol (2003)*, "On the explicit formula" (Hilbert-space approach, Weil positivity).
- *Connes (1998)*, "Trace formula in RH context" (operator-algebra approach).
- *Deninger (1992)*, "Analogies between number theory and dynamics" (spectral analogy).
- *Baez-Duarte (2005)*, "Strengthening of the Nyman–Beurling criterion" (closure theorem).
- *Suzuki (2015)*, "Li's coefficients as norms of functions" (functional model for RH).
- *LMFDB, Platt–Trudgian, Arb, mpmath documentation* for data and tools.

These provide context and rigor behind each element of the NEXUS program. No known result in them completes RH; rather, they form the scaffold into which our lemmas and numerics must fit.

Summary: In NEXUS terms, we have two complementary "domains" (arithmetic and analytic) and a "nexus" (the mirror). The RH statement becomes "the cycle anchored at the nexus has no drift off the seam." All steps in this report tie back to that principle. The RF-target is explicit, and the path to it is broken into verifiable lemmas and computational checks. At this point, the research program is clearly laid out: the remaining open question is proving the positivity of the construction $\mathcal{Q}_{\mathrm{prime}}$ against phase interference (the so-called closure of the cycle). All other elements have been formulated and (partially) tested numerically.

NEXUS-RH as a Runtime-Safety Research Program

Executive summary

The safest rigorous interpretation of the NEXUS-RH program is not "RH is already proved," but "RH can be recast as an operator-theoretic runtime-safety target whose proof obligations are explicit, testable, and numerically auditable." In that recast, Gate A is a compile-time diagnostic layer based on Jensen-polynomial hyperbolicity and Laguerre–Pólya heuristics; Gate B is the runtime layer, where an arithmetic cascade is coupled to the analytic mirror from the functional equation. The proof target becomes the absence of illegal off-seam states: either $\ker(I + \mathbb{L}_s) = \{0\}$ for $\Re(s) > 1/2$, or, more naturally by Schur reduction, $1 \notin \mathrm{Spec}(R_s)$ for the round-trip operator R_s. This is a genuine research program, not an already accepted proof; RH remains open.

A rigorous version of the framework should be built on weighted Mellin/log-L^2 spaces, a renormalized Buchstab/Volterra operator K_s^{ren}, and a bounded mirror $J_R(s)$ induced by inversion $x \mapsto 1/x$ plus the functional-equation multiplier $\chi(s)$. The doubled bundle $\mathbb{H}_s = H_s \oplus H_{1-s}$ then carries an off-diagonal loop operator

$$\mathbb{L}_s = \begin{pmatrix} 0 & J_R(s)K_{1-s}^{\mathrm{ren}} \\ J_R(1-s)K_s^{\mathrm{ren}} & 0 \end{pmatrix}.$$

If each one-fiber operator is Hilbert–Schmidt, then the one-step loop need only be 2-determinant class, while the Schur-reduced round-trip

$$R_s := J_R(s)K_{1-s}^{\mathrm{ren}}J_R(1-s)K_s^{\mathrm{ren}}$$

is trace class, so $\det(I - R_s)$ is well-defined and becomes the mathematically natural Gate-B observable. This is exactly the point where Bornemann-style Nyström numerics become relevant.

The de Bruijn–Newman heat flow gives the right external calibration for the framework's "fold-pressure" parameter λ: de Bruijn and Newman showed that the heat-deformed ξ-family has a threshold constant Λ, RH is equivalent to $\Lambda \leq 0$, Rodgers and Tao proved $\Lambda \geq 0$, and Polymath improved the unconditional upper bound to $\Lambda \leq 0.22$. This makes λ an objective calibration axis for Gate B rather than a metaphor. A candidate internal constant H such as $\pi/9$ should therefore be treated only as a calibration hypothesis against pre-registered observables R_1, R_2, R_3, never assumed as an identity.

The report below gives a rigorous architecture, three bridge lemmas, a heat-calibrated Gate-B family $\mathbb{L}_{s,\lambda}$, and a reproducible numerical program using LMFDB zeros, Platt–Trudgian verification windows, segmented sieve data, Arb ball arithmetic, and Bornemann-style Fredholm numerics. It also makes one strategic recommendation: the main proof target should be the Schur-reduced round trip R_s, not raw one-step norms, because R_s is the determinant-class object most closely aligned with runtime safety, explicit-formula positivity, Li/Weil closure functionals, and Nyman–Beurling-type closure criteria.

NEXUS framing and operator architecture

NEXUS is best used here as an **interpretive lens**: a *domain* is a Hilbert or Banach state space together with its observable algebra and admissible read/write operators, while a *nexus* is a bounded intertwiner or return map coupling two domains. For

RH, the arithmetic domain is the signed rough-integer or first-prime-break side; the analytic domain is the ζ/ξ reflection side; the nexus is the two-fiber mirror. This framing is compatible with existing operator antecedents: Burnol's conductor/operator reading of the explicit formula, Connes's trace-formula spectral picture, Deninger's dynamical/trace analogy, de Branges-type Hilbert-space programs, Suzuki's model-space norm formulations of Li coefficients, and Báez-Duarte's closure-theoretic Nyman–Beurling criterion.

A clean formal starting point is the half-density Mellin space

$$H_s := L^2\left(\mathbb{R}_+,\, x^{2\sigma-1}\rho_\eta(\log x)\,\frac{dx}{x}\right), \qquad s=\sigma+it,\ \sigma\in(0,1),$$

with $\rho_\eta(u)=e^{-2\eta|u|}$ as the default log-window and $\eta=1/2$ as the prototype choice. The conjugacy

$$(U_s f)(u) := e^{\left(\sigma-\frac{1}{2}\right)u}\rho_\eta(u)^{1/2} f(e^u), \qquad u=\log x,$$

identifies H_s with ordinary $L^2(\mathbb{R}, du)$. This is the right place to build Mellin-convolution/Volterra kernels. A prime-density factor such as $\left(\log(e+x)\right)^{-1}$ is a legitimate alternative modeling choice, but it interacts badly with exact inversion symmetry; the clean default is therefore to keep prime-density corrections inside the kernel rather than in the norm. That is an open choice, not a theorem.

Define a renormalized Buchstab leg by a log-Volterra kernel. Let $\kappa^{\mathrm{ren}}: [0,\infty)\to\mathbb{C}$ be the renormalized first-prime-break profile extracted from Buchstab recursion after subtraction of the chosen main term and the designated boundary residue. Then

$$(K_s^{\mathrm{ren}} f)(x) := \int_1^x \kappa_s^{\mathrm{ren}}\left(\log(x/y)\right) f(y)\,\frac{dy}{y}, \qquad \kappa_s^{\mathrm{ren}}(\tau) := e^{-s\tau}\kappa^{\mathrm{ren}}(\tau),\ \tau\ge 0.$$

This is the precise place where the program must "compile" arithmetic residue data into an operator. The Buchstab side is justified by the classical identity for $\Phi(x,y)$, its delay equation, and its role as the universal first-prime-break mechanism for rough counts.

The mirror leg should be defined from the functional equation. DLMF records the ξ-symmetry $\xi(s)=\xi(1-s)$ and the standard functional-equation multiplier $\chi(s)$ for $\zeta(s)$. A bounded model is

$$(J_R(s) f)(x) := j_R(s)\, E_R(x;s)\, f(1/x),$$

where $E_R(\cdot;s)$ is an optional bounded Euler dressing and $j_R(s)$ is a scalar chosen so that $j_R(s)j_R(1-s)=1$. The natural default is $j_R(s)=\chi(s)^{-1/2}$ with a consistent branch choice. If $E_R\equiv 1$ and ρ_η is even, inversion defines a bounded map $H_{1-s}\to H_s$; on the critical line, where $|\chi(1/2+it)|=1$, the mirror is unitary up to the chosen branch and bounded dressing.

Now double the space:

$$\mathbb{H}_s := H_s \oplus H_{1-s}, \qquad \mathbb{K}_s := \begin{pmatrix} 0 & K_{1-s}^{\mathrm{ren}} \\ K_s^{\mathrm{ren}} & 0 \end{pmatrix}, \qquad \mathbb{J}_R(s) := \begin{pmatrix} 0 & J_R(s) \\ J_R(1-s) & 0 \end{pmatrix}.$$

The actual Gate-B loop should be defined directly as

$$\mathbb{L}_s := \begin{pmatrix} 0 & J_R(s)K_{1-s}^{\mathrm{ren}} \\ J_R(1-s)K_s^{\mathrm{ren}} & 0 \end{pmatrix}.$$

Its Schur-reduced round trip on the s-fiber is

$$R_s := J_R(s)K_{1-s}^{\mathrm{ren}} J_R(1-s) K_s^{\mathrm{ren}}: H_s \to H_s.$$

The explicit proof targets are

$$\ker(I + \mathbb{L}_s) = \{0\}, \qquad \| \mathbb{L}_s \| < 1 \quad \text{for } \Re(s) > 1/2.$$

The second target is stronger than necessary. The mathematically natural runtime-safety target is

$$1 \notin \operatorname{Spec}(R_s),$$

or the sufficient contraction estimate $\| R_s \| < 1$; by Schur complement, that already forces $\ker(I + \mathbb{L}_s) = \{0\}$. This is the right place to place the main proof burden.

The clean operator-class conditions are these. If the conjugated kernel $\tilde{k}_s$ of $U_s K_s^{\mathrm{ren}} U_s^{-1}$ belongs to $L^2(\mathbb{R}^2)$, then $K_s^{\mathrm{ren}} \in \mathcal{S}_2$ and is compact. If $\tilde{k}_s$ factors as

$$\tilde{k}_s(u, v) = \int_{\mathbb{R}} a_s\,(u, w) b_s(w, v)\, dw$$

with $a_s, b_s \in L^2(\mathbb{R}^2)$, then $K_s^{\mathrm{ren}} = A_s B_s \in \mathcal{S}_1$. More generally, sharp kernel conditions for Schatten classes are available from Delgado–Ruzhansky. In the doubled setting, a very useful structural fact is that if both one-fiber legs are Hilbert–Schmidt and the mirror is bounded, then each off-diagonal block of $\mathbb{L}_s$ is Hilbert–Schmidt, while the round trips R_s and R_{1-s} are trace class because products of two Hilbert–Schmidt operators lie in $\mathcal{S}_1$. Therefore $\det(I - R_s)$ is well-defined even when $\mathbb{L}_s$ itself only supports the modified determinant $\det_2(I + \mathbb{L}_s)$. That observation is central to making Gate B numerically and analytically viable.

The main unspecified choices, together with reasonable defaults, are these:

Modeling choice	Status	Default	Sensitivity sweep
Log-window ρ_η	open	$\eta = 0.5$	0.25,0.5,0.75
Prime density in norm vs kernel	open	keep in kernel	compare with $dx/\big(x\log(e + x)\big)$
Buchstab renormalization main term	open	subtract $e^{-\gamma}$-level asymptotic after smoothing	compare local polynomial or spline main term
Euler dressing E_R	open	$E_R \equiv 1$	finite Euler product $p \leq 29{,}43{,}61$
Physical x-range	open	$x_{\max} = 10^6$ prototype	$10^7, 10^8$ if memory allows
Basis	open	symmetrized log-Laguerre	log-wavelets, Chebyshev-on-window
Determinant	conditional	$\det(I - R_s)$	compare $\det_2(I + \mathbb{L}_s)$
Floating precision	open	10^{-12}	Arb-certified 10^{-30}

Bridge lemmas and analytic route

The first bridge lemma should convert arithmetic residue data into an operator coefficient system.

Lemma 1. *Suppose an arithmetic residue observable $I(x)$ admits a smoothed first-prime-break decomposition*

$$I_W(X; s) := \int_1^\infty I\,(x)\, W(x/X)\, x^{-s}\, \frac{dx}{x}$$

with $W \in C_c^\infty(\mathbb{R}_+)$, and suppose the decomposition can be written as a renormalized Buchstab identity with signed breakpoint profile κ^{ren} plus a boundary term $B_W(X;s)$ satisfying $B_W(X;s) = o_X(1)$ uniformly on compact s-sets. Then there exists a Volterra Mellin operator $K_{s,X}^{\mathrm{ren}}$ such that for every finite basis $\{\phi_n\}_{n\le N} \subset H_s$,

$$I_W(X;s) = \sum_{m,n\le N} c_{mn}(s;X)\,\langle \phi_m, K_{s,X}^{\mathrm{ren}}\phi_n\rangle + E_{N,X}(s),$$

with $E_{N,X}(s) \to 0$ as $N \to \infty$ and $X \to \infty$ along the chosen smoothing regime. If $\tilde{k}_{s,X} \in L^2(\mathbb{R}^2)$ uniformly in X, then $K_{s,X}^{\mathrm{ren}}$ is uniformly Hilbert–Schmidt and every trace-class statement can be pushed to the round trip $R_{s,X}$.

The proof strategy is standard: start from the Buchstab identity

$$\Phi(x,y) = \Phi(x,z) + \sum_{y<p\le z}\sum_{v\ge 1}\Phi\left(x/p^v, p\right),$$

pass to logarithmic variables so that first-prime-factor recursion becomes a delayed or Volterra-type convolution, then apply Mellin smoothing to isolate operator coefficients. Fan's explicit rough-number estimates are useful for error management, while Lagarias's sieve survey is the right conceptual background. Schatten control comes from the L^2-kernel condition after conjugation.

The second bridge lemma is the structural core of Gate B.

Lemma 2. *Let*

$$A_s := J_R(s)K_{1-s}^{\mathrm{ren}}, \qquad B_s := J_R(1-s)K_s^{\mathrm{ren}}, \qquad \mathbb{L}_s = \begin{pmatrix} 0 & A_s \\ B_s & 0 \end{pmatrix}.$$

If A_s, B_s are bounded on $H_{1-s} \to H_s$ and $H_s \to H_{1-s}$, then

$$I + \mathbb{L}_s \text{ is invertible } \Leftrightarrow I - A_sB_s \text{ and } I - B_sA_s \text{ are invertible.}$$

If $A_s, B_s \in \mathcal{S}_2$, then $A_sB_s, B_sA_s \in \mathcal{S}_1$, and

$$\det(I - A_sB_s) = \det(I - B_sA_s)$$

is well-defined. Consequently, the runtime-safety problem may be reduced from the doubled bundle to the one-fiber round trip $R_s = A_sB_s$.

The proof is the usual Schur-complement argument. This is the rigorous reason to make R_s, not $\mathbb{L}_s$, the primary target. It also explains why a Hilbert–Schmidt proof on each fiber is enough to reach a trace-class determinant after one complete mirror-cascade cycle. This is the right place to connect with Bornemann numerics and modified-determinant theory.

The third bridge lemma is the exclusion principle.

Lemma 3. *Assume R_s is analytic in s on $\{\Re(s) > 1/2\}$ as an $\mathcal{S}_1$-valued map, and let P_N denote a finite-section projector for a spectrally complete basis. Suppose $R_s^{(N)} := P_NR_sP_N \to R_s$ in trace norm uniformly on compact s-sets. If there exists $\delta > 0$ such that*

$$s_{\min}\left(I - R_s^{(N)}\right) \ge \delta$$

for all large N and all s in a compact subset of $\{\Re(s) > 1/2\}$, then

$$1 \notin \mathrm{Spec}(R_s), \qquad \ker(I + \mathbb{L}_s) = \{0\}.$$

If, in addition, the finite-section characteristic polynomials are generated by stability-preserving truncation maps, Borcea–Brändén theory can be used to rule out spurious nonreal crossings introduced by the discretization.

The proof strategy uses the finite-section method plus continuity of singular values and trace-norm convergence of Fredholm determinants. The Borcea–Brändén component is optional and conditional: it applies only if the particular discretization/basis map is shown to preserve the relevant stability class. That is a real theorem route, not a slogan.

Three exact closure-function routes then become visible. A **Weil route** seeks a Gate-B quadratic form that is a concrete representative of Weil's positivity criterion; a **Li route** seeks a norm model analogous to Suzuki's "Li coefficients as norms"; and a **closure/approximation route** seeks an explicit Gate-B representative of Báez-Duarte–Nyman–Beurling closure. These are all mathematically exact RH-equivalent criteria already known in the literature; the NEXUS task is not to replace them with metaphor, but to build a *specific* operator $\mathcal{Q}_{\text{prime}}$ inside Gate B that lands inside one of those exact frameworks.

Heat calibration and Gate diagnostics

The de Bruijn–Newman family is the correct external calibration axis for NEXUS "fold-pressure." De Bruijn introduced heat-deformed Fourier transforms with a real-zero threshold, Newman proved the existence of a finite threshold constant Λ, RH is equivalent to $\Lambda \leq 0$, Rodgers and Tao proved $\Lambda \geq 0$, and Polymath 15 improved the unconditional upper bound to $\Lambda \leq 0.22$. Any NEXUS parameter λ that purports to encode fold-pressure should therefore be coupled directly to a heat deformation of the Gate-B kernel rather than treated as an abstract free constant.

A mathematically clean proposal is to heat only the difference variable of the conjugated log-kernel:

$$\tilde{k}_{s,\lambda}(u,v) := \left(e^{\lambda\,\partial_\tau^2}\tilde{\kappa}_s(\tau)\right)|_{\tau=u-v}\, a_\eta(u,v),$$

and then define $K_{s,\lambda}^{\text{ren}}$ from $\tilde{k}_{s,\lambda}$ and

$$\mathbb{L}_{s,\lambda} := \begin{pmatrix} 0 & J_R(s)K_{1-s,\lambda}^{\text{ren}} \\ J_R(1-s)K_{s,\lambda}^{\text{ren}} & 0 \end{pmatrix}.$$

This is the direct Gate-B analog of the de Bruijn–Newman heat flow. It keeps the mirror fixed while varying the arithmetic leg by a controllable heat parameter. That design is preferable to heating the mirror multiplier itself, because the classical heat flow acts on the Fourier/Mellin side of the kernel, not on the involution.

Gate A should be kept, but only as a compile-time diagnostic. The clean version is the Jensen–Pólya program: define the relevant coefficient sequence from the even Taylor expansion of $\Lambda(1/2+z)$, build Jensen polynomials $J_{d,n}$, and test hyperbolicity. Griffin, Ono, Rolen, and Zagier proved asymptotic hyperbolicity for a density-1 subset in each fixed degree and established large finite ranges; but Farmer has made a substantial case that this route, by itself, is not plausibly strong enough to prove RH. In NEXUS terms, Gate A is a type check, not a runtime proof.

A candidate internal constant H, including the frequently proposed $H = \pi/9$, should be treated as a **calibration hypothesis** against the following observables, none of which assumes equality in advance:

Observable	Definition	Interpretation
$R_1(\sigma,t;N)$	$\dfrac{s_{\min}\left(I - R_s^{(N)}\right)}{\sigma - \frac{1}{2}}$	normalized spectral gap as seam distance closes
$R_2(\sigma,t;N)$	$-\partial_\lambda \log\left\|\det\left(I - R_{s,\lambda}^{(N)}\right)\right\| \|_{\lambda=0}$	heat susceptibility of the trace-class loop

Observable	Definition	Interpretation
$R_3(\sigma, t; N, M)$	empirical slope of $\log \parallel R_s^{(N)\,m} v_0 \parallel$ for $1 \le m \le M$	Neumann decay/contraction rate

The rule for a serious calibration claim is strict: pre-register the normalization, run independent bases and windows, compare to H only after error bars are fixed, and require consistency across N, $x_{\max}$, η, and basis family. Anything weaker is numerology, not calibration. This report therefore treats $H = \pi/9$ as a testable *output* of the program, not as an input.

Numerical program and reproducibility

The public data side is mature enough to support real experiments. LMFDB currently serves the first 103,800,788,359 zeros of $\zeta(s)$ above the real axis, all listed with real part $1/2$ and imaginary parts stored to absolute precision $\pm 2.5 \times 10^{-31}$. Independently, Platt and Trudgian rigorously verified RH up to height $3 \cdot 10^{12}$ using interval arithmetic. Arb provides ball arithmetic for rigorous certification, while mpmath remains useful for high-level prototyping.

The minimum viable reproducible pipeline is:

```
flowchart TD
  A[Segmented sieve to x_max] --> B[Compute primes, LPF/SPF, rough counts Φ(x,y)]
  B --> C[Renormalize first-prime-break profile κ^ren]
  C --> D[Build weighted log-space kernel K_s^ren]
  D --> E[Build mirror J_R(s) from inversion plus χ(s) and optional E_R]
  E --> F[Assemble doubled loop L_s and round trip R_s]
  F --> G[Compute spectra, s_min(I-R_s), ||R_s||, det(I-R_s)]
  G --> H[Heat scan λ ↦ R_{s,λ}]
  H --> I[Calibrate R1,R2,R3 and certify with Arb]
```

The implementation should not start by sampling the full critical strip. It should start with a **trusted stair-step**: fixed $t \in \{0, 14.1347, 21.0220, 25.0109\}$, $\sigma \in \{0.75, 0.70, \dots, 0.55, 0.525, 0.51\}$, $x_{\max} = 10^6$, symmetrized log-Laguerre basis sizes $M = 64, 128, 256$, $U = \log x_{\max}$, and $\eta = 0.5$. Once convergence and basis stability are visible there, extend to $x_{\max} = 10^7$ or 10^8, then introduce λ-scans. The first acceptance criterion is not "norm less than one everywhere," but basis- and truncation-stable positivity of $s_{\min}(I - R_s)$.

A reproducible pseudocode sketch is:

```
# Stage A: arithmetic data
primes, spf = segmented_sieve_with_spf(x_max)
Phi = rough_count_table(x_grid, y_grid, spf)        # Φ(x,y)
kappa_ren = renormalize_first_prime_break(Phi, method="smoothed_main_term")

# Stage B: basis and kernels
basis = sym_log_laguerre_basis(M, U=np.log(x_max), eta=0.5)
def K_matrix(s, kappa_ren, basis):
  return project_volterra_mellin_kernel(s, kappa_ren, basis)

Ks   = K_matrix(s,     kappa_ren, basis)
K1s  = K_matrix(1 - s, kappa_ren, basis)
```

```
# Stage C: mirror
chi_s = zeta_functional_multiplier(s)          # from chi(s)
JR_s  = mirror_matrix(s, basis, chi_s, euler_dress=None)
JR_1s = mirror_matrix(1 - s, basis, zeta_functional_multiplier(1-s))

# Stage D: bundled loop and round trip
A = JR_s  @ K1s
B = JR_1s @ Ks
L = np.block([[0*A, A],
        [B,   0*B]])
R = A @ B

# Stage E: diagnostics
eig_R  = eigvals(R)
smin_R = smallest_singular_value(np.eye(R.shape[0]) - R)
norm_R = spectral_norm(R)
det_R  = fredholm_det_trace_class(R)          # or det(I-R) in matrix truncation
neumann = [norm(matrix_power(R, m) @ v0) for m in range(1, M_iter+1)]
```

To map an arithmetic interior residue $I(x)$ into operator coefficients, the clean route is to work with smoothed Mellin moments:

$$\mathcal{M}_I(s;X) = \int_1^\infty I(x)\, W(x/X)\, x^{-s} \frac{dx}{x},$$

then write

$$\mathcal{M}_I(s;X) \approx \sum_{m,n\le M} \hat{I}_{mn}(s;X)\, \langle \phi_m, K_s^{\mathrm{ren}} \phi_n \rangle.$$

The coefficients $\hat{I}_{mn}$ can be learned from the smoothed rough-count decomposition or, for cross-checking, from direct quadrature against basis test functions. This is exactly where Mellin and Laplace transforms become bridge mechanics rather than metaphors.

The recommended parameter grid is:

Component	Prototype default	Sensitivity values	Notes
$x_{\max}$	10^6	$10^7, 10^8$	exact SPF feasible at 10^6; segmented only above
log-window η	0.5	0.25,0.75	controls HS regularization
basis family	sym log-Laguerre	log-wavelet, Chebyshev-windowed	require basis-independence checks

Component	Prototype default	Sensitivity values	Notes
basis size M	256	$64,128,512$	use doubling to test finite-section convergence
s-grid	$\sigma \in [0.51,0.80]$, selected t	denser near $1/2$	include first few zero ordinates and off-resonant t
λ-grid	0 and $\pm 10^{-4}, \pm 5 \cdot 10^{-4}, \pm 10^{-3}$	up to 10^{-2} if stable	heat susceptibility should be local first
quadrature	Gauss–Legendre/Clenshaw–Curtis	adaptive composite Simpson	Bornemann-compatible choices
float tolerance	10^{-12}	10^{-14}	prototype only
rigorous tolerance	Arb radius 10^{-30}	10^{-40}	certify decisive runs

A rigorous task schedule is:

```
timeline
  title Gate-B research sequence
  Architecture lock : Define H_s, J_R(s), K_s^ren, L_s, R_s
  Arithmetic compile : Implement sieve/SPF, rough counts, Buchstab renormalization
  One-fiber numerics : Project K_s^ren on symmetrized log bases
  Bundle numerics : Assemble A, B, L_s, R_s and scan s-grid
  Heat calibration : Build λ-family and measure R1, R2, R3
  Certification : Repeat key runs with Arb and determinant error bounds
  Analytic closure : Prove Schur-reduced exclusion and align with Weil/Li/Nyman
```

Because no validated experiment output accompanies this report, the correct way to "include charts" is to provide plotting scripts that consume real CSV outputs rather than invent numbers. The minimal CSV schema is **[sigma, t, lambda, basis, M, x_max, eta, eig_real, eig_imag, norm_R, smin_I_minus_R, det_I_minus_R, neumann_step, neumann_norm, certified_radius]**. Then:

```python
# Spectra vs Re(s)
import pandas as pd, matplotlib.pyplot as plt
df = pd.read_csv("gateB_scan.csv")
for tval, g in df.groupby("t"):
  plt.figure()
  plt.scatter(g["sigma"], g["eig_real"], s=8)
  plt.xlabel("Re(s)")
  plt.ylabel("Re eigenvalues of R_s")
  plt.title(f"Round-trip spectrum slices at t={tval}")
  plt.show()

# Norm vs Re(s)
for tval, g in df.groupby("t"):
```

```
  h = g.sort_values("sigma")
  plt.figure()
  plt.plot(h["sigma"], h["norm_R"])
  plt.plot(h["sigma"], h["smin_I_minus_R"])
  plt.xlabel("Re(s)")
  plt.ylabel("norm / singular value")
  plt.title(f"Contraction diagnostics at t={tval}")
  plt.legend(["||R_s||", "s_min(I-R_s)"])
  plt.show()

# Neumann decay curves
nd = pd.read_csv("neumann_decay.csv")
for key, g in nd.groupby(["sigma", "t", "basis", "M"]):
  h = g.sort_values("neumann_step")
  plt.figure()
  plt.semilogy(h["neumann_step"], h["neumann_norm"])
  plt.xlabel("iteration m")
  plt.ylabel("||R_s^m vo||")
  plt.title(f"Neumann decay for sigma={key[0]}, t={key[1]}, basis={key[2]}, M={key[3]}")
  plt.show()
```

Deliverables and prioritized bibliography

The concrete deliverables should be four fold. The first is a short **operator note** fixing the definitions of H_s, K_s^{ren}, $J_R(s)$, $\mathbb{L}_s$, and R_s, together with the exact Schur-complement reduction and the precise open modeling choices. The second is a **numerical paper** reporting finite-section spectra, $s_{\min}(I - R_s)$, determinant stability, and λ-susceptibilities with fully reproducible code and Arb certification. The third is an **analytic bridge note** proving Lemmas 1–3 under explicit hypotheses and identifying which hypothesis is still unproved. The fourth is a **closure-functional paper** whose only purpose is to connect R_s or an allied quadratic form to one exact RH-equivalent criterion: Weil positivity, Li positivity, or Nyman–Beurling closure. That last paper is the actual proof bottleneck.

The prioritized public sources should be read in this order. For the heat side: de Bruijn 1950, Newman 1976, Rodgers–Tao 2018/2021, and Polymath 15. For arithmetic decomposition: Buchstab's classical mechanism via modern explicit references such as Fan and Lagarias. For operator/explicit-formula antecedents: Burnol, Connes, Deninger, Burnol's Sonine-space paper, and Báez-Duarte. For Gate A: Griffin–Ono–Rolen–Zagier, plus Farmer's critique. For numerics: Bornemann, Gesztesy–Makarov for 2-modified determinants, LMFDB, Platt–Trudgian, Arb, and mpmath.

The key bibliography, in priority order, is:

Priority	Source	Why it matters
highest	de Bruijn, *The roots of trigonometric integrals*	existence of the heat threshold and the original real-zero deformation
highest	Newman, *Fourier transforms with only real zeros*	existence of Λ and RH $\Leftrightarrow \Lambda \le 0$
highest	Rodgers–Tao, *The de Bruijn–Newman*	$\Lambda \ge 0$

Priority	Source	Why it matters
	constant is non-negative	
highest	Polymath 15, *Effective approximation of heat flow evolution…*	unconditional $\Lambda \leq 0.22$
highest	Burnol, *The Explicit Formula in simple terms*	Weil positivity, additive/multiplicative convolution, conductor operator
highest	Báez-Duarte, *A strengthening of the Nyman–Beurling criterion*	exact closure criterion
high	Suzuki, *Li coefficients as norms of functions in a model space*	norm-functional form directly relevant to $\mathcal{Q}_{\mathrm{prime}}$
high	Fan, *Numerically explicit estimates for the distribution of rough numbers*	Buchstab identity and explicit rough-number control
high	Bornemann, *On the Numerical Evaluation of Fredholm Determinants*	Nyström error control and practical determinant numerics
high	Gesztesy–Makarov, *Modified Fredholm Determinants…*	2-modified determinants for matrix/operator systems
high	Griffin–Ono–Rolen–Zagier	Gate-A compile-time diagnostic via Jensen hyperbolicity
high	Farmer	caution that Gate A alone is not a plausible proof path
medium	Connes	spectral/trace-formula view of critical zeros vs resonances
medium	Deninger	explicit-formula/dynamical-system analogy
medium	Burnol, Sonine-space paper	quotient vectors attached to zeros
medium	LMFDB, Platt–Trudgian, Arb, mpmath	public numerical data and certification stack

The final methodological recommendation is simple. In NEXUS language, RH should be pursued as a **two-domain closure problem**: arithmetic cascade plus analytic mirror, with the nexus embodied by a bounded return operator. In standard mathematical language, that means: define K_s^{ren} rigorously enough to prove Hilbert–Schmidt regularity, define $J_R(s)$ rigorously enough to control the mirror, reduce the doubled bundle to the trace-class round trip R_s, and then identify or construct a prime-field quadratic form that makes $1 \in \mathrm{Spec}(R_s)$ impossible for $\Re(s) > 1/2$. That is the point where "runtime safety" becomes theorem rather than metaphor.

Universal Action Type System and Prime-Field Closure

Executive summary

The project's internal synthesis can be put into rigorous mathematical form if one stops treating "query," "signature," and "closure" as metaphors and instead treats them as operator-theoretic data: a symmetry representation, a spectral aperture, a signed kernel, a decomposition into annihilated and surviving channels, a scale law, and a positive quadratic closure form. In that language, the project's informal rule "glyph → operation → signature → compatible invariant → readout" becomes a precise statement about bounded operators between Hilbert or representation spaces. The "Universal Action Type System" is therefore best formalized as a typed calculus of symmetry-compatible readout operators, not as a replacement for classical analysis.

The BBP formula is the canonical prototype. In the original paper and in Bailey's later compendium, BBP is a base-16 digit-extraction identity for π; but when rewritten through its integral representation, it becomes a clean example of a query operator whose signed residue mask annihilates one channel exactly and leaves a phase channel as the only surviving boundary residue. That is not philosophical language: the cancellation can be derived explicitly from the BBP series via a Maclaurin expansion and a Mellin-moment calculation, followed by a rational partial-fraction decomposition.

On the RH side, one part of the project is already mathematically solid: the single-family detector

$$Q_{\omega}(\alpha,\gamma) = \int \left|D_{\alpha,\gamma}(u)\right|^{2} \omega(u)\, du$$

is a genuine positive quadratic functional, and with mild assumptions on ω one can prove

$$Q_{\omega}(\alpha,\gamma) = 0 \Leftrightarrow \alpha = 0.$$

This is a local statement about one zero family and it does not require RH. The internal project notes correctly separate this local detector from the still-open global "prime-field closure functional" Q_{prime}.

A crucial correction is also now clear. The prime number theorem by itself does **not** rule out off-line zeros. In the explicit formula, a zero $\rho = \beta + i\gamma$ contributes roughly x^{β}/ρ, and if $\beta < 1$ then this is still $o(x)$, so $\psi(x) \sim x$ remains compatible with such a term. What must be built is not a PNT contradiction, but an RH-equivalent positive form that detects and forbids seam-orthogonal defect energy. That is why the earlier "domination lemma" direction was too strong and why the correct target is a closure functional with a no-cross-phase-escape theorem.

Among the classical RH-equivalent criteria, the closest candidates for the sought Q_{prime} are the Weil quadratic functional and the Nyman–Beurling closure norm. Bombieri's study of the Weil form is especially relevant because he proves that the associated quadratic functional is positive semidefinite iff RH, and that if RH fails only through finitely many off-line zeros, then large truncations develop exactly half as many negative eigenvalues as offending zeros. Li's coefficients are important, but they are best viewed as a discrete moment-sampling of Weil's form rather than as a naturally local detector basis. Nyman–Beurling is already a bona fide Hilbert-space closure norm, so it is the cleanest "closure" object, but it does not by itself diagonalize the project's local defect packets.

The mathematically correct "mold" for the open problem is therefore this: construct a prime-side Hilbert-space norm in which the seam-orthogonal defect packets form a **Riesz sequence** or at least have a uniform lower frame bound. In symbols, one wants a functional Q_{prime} with

$$Q_{\text{prime}}\left(\sum_{\rho} a_\rho D_\rho\right) \geq A \sum_{\rho} |a_\rho|^2 Q_\omega(\alpha_\rho, \gamma_\rho), \qquad A > 0,$$

for all finite coefficient families. That is the natural precise meaning of "no cross-phase escape." It converts the user's shape-language into a standard problem in positive quadratic forms, Mellin localization, and frame theory, and it is the right frontier to pursue. The report below develops that program carefully, marking which parts are classical, which are rigorous new formalizations, and which are still conjectural.

Signature calculus

Because "Universal Action Type System" and $\Sigma(Q) \sim \Sigma(I)$ are not standard published terms, the correct way to proceed is to define them as a formal layer built on standard harmonic-analysis objects. An intermediate project memo had already compressed signatures to a shorter parity tuple; the refinement below makes the channel decomposition and closure functional explicit, which is what the global RH program needs.

Let X denote either a **query** Q or an **invariant** I. Fix a separable Hilbert space H_X, a locally compact symmetry group or semigroup G_X, and a strongly continuous representation $\pi_X: G_X \to U(H_X)$. Define the signature of X by

$$\Sigma(X) = (\mathcal{S}_X, \mathcal{A}_X, \mathcal{K}_X, \mathcal{E}_X, \mathcal{R}_X, \mathcal{B}_X, \mathcal{C}_X),$$

with the following components.

Component	Formal definition	Intended meaning
$\mathcal{S}_X$	symmetry datum (G_X, π_X, J_X), with J_X a distinguished involution or mirror operation when present	symmetry/parity law
$\mathcal{A}_X$	measurable subset of spectral space or orbit space for π_X	address aperture
$\mathcal{K}_X$	bounded operator, distribution, or signed finite measure supported on $\mathcal{A}_X$	kernel / weight mask
$\mathcal{E}_X$	closed J_X-stable subspace of H_X	exhaust channel
$\mathcal{R}_X$	closed complementary J_X-stable subspace, with $H_X = \mathcal{E}_X \oplus \mathcal{R}_X$	residue channel
$\mathcal{B}_X$	dilation or scale semigroup D_t commuting with the symmetry action	base / scale law
$\mathcal{C}_X$	positive semidefinite quadratic form $f \mapsto \langle M_X f, f\rangle$, $M_X \geq 0$	closure rule

The word **parity** is made precise by the involution J_X: the signature remembers which parts of the data are J_X-even, J_X-odd, or more generally annihilated versus preserved under readout. This is the mathematically clean way to recover the project's "shape-channel" language. The closure rule is required to be quadratic because every viable global RH criterion in the literature that resembles the project's target—Weil, Li, Nyman–Beurling—ultimately becomes a positive or sign-controlled quadratic form.

The compatibility relation $\Sigma(Q) \sim \Sigma(I)$ should then be defined by the existence of a bounded intertwiner $T: H_I \to H_Q$ satisfying five conditions:

$$T\pi_I(g) = \pi_Q(\varphi(g))T,$$

for an isomorphism or embedding of the relevant symmetry data; the active spectral support of TI lies inside $\mathcal{A}_Q$; the query kernel sends invariant exhaust into query exhaust and kills it at closure,

$$\mathcal{C}_Q(\mathcal{K}_Q Te) = 0 \quad \text{for all } e \in \mathcal{E}_I;$$

the residue map $P_{\mathcal{R}_Q}\mathcal{K}_Q T|_{\mathcal{R}_I}$ is nonzero, ideally injective; and the two scale laws commute after reparametrization,

$$TD_t^{(I)} = D_{\psi(t)}^{(Q)}T.$$

This says in standard language what the project means by "the query geometry resonates with the invariant." It is not mystical: it is an intertwining-and-projection condition.

A useful elementary proposition falls right out of this formalism.

Proposition. If $H_I = \mathcal{E}_I \oplus \mathcal{R}_I$, $H_Q = \mathcal{E}_Q \oplus \mathcal{R}_Q$, and $K := \mathcal{K}_Q T$ satisfies $P_{\mathcal{R}_Q}K|_{\mathcal{E}_I} = 0$ and $P_{\mathcal{R}_Q}K|_{\mathcal{R}_I}$ injective, then the readout operator $P_{\mathcal{R}_Q}K$ factors through the quotient $H_I/\mathcal{E}_I$, and vanishing readout implies vanishing invariant residue.

Proof. Write $v = e + r$ with $e \in \mathcal{E}_I$, $r \in \mathcal{R}_I$. Then

$$P_{\mathcal{R}_Q}K(v) = P_{\mathcal{R}_Q}K(e) + P_{\mathcal{R}_Q}K(r) = P_{\mathcal{R}_Q}K(r),$$

so the readout depends only on the residue class $[v] \in H_I/\mathcal{E}_I$. If $P_{\mathcal{R}_Q}K(v) = 0$, then $P_{\mathcal{R}_Q}K(r) = 0$, and injectivity on $\mathcal{R}_I$ forces $r = 0$. ■

That proposition captures the core project heuristic in rigorous form: "closure" means that the exhaust channel is quotiented out, and valid readout lives only on the surviving residue class. The same idea is visible in the classical Mellin-side explicit formula, where primes and zeros are linked by a transform that can be organized into a symmetry-compatible quadratic form.

```
flowchart LR
  I[Invariant state in H_I] --> T[Intertwiner T]
  T --> A[Address aperture A_Q]
  A --> K[Kernel / weight mask K_Q]
  K --> E[Exhaust channel]
  K --> R[Residue channel]
  E --> C[Closure rule C_Q = 0]
  R --> O[Readout on residue quotient]
```

Two examples are enough to anchor the formalism. For BBP, the symmetry is the residue wheel modulo 8 together with base-16 dilation; the aperture is the residue subset $\{1,4,5,6\}$; the kernel is the signed mask $(4,0,0,-2,-1,-1,0,0)$; the exhaust channel is logarithmic/radial boundary data; the residue channel is angular/arctangent phase; the scale law is $k \mapsto 16^{-k}$; and the closure rule is exact vanishing of the radial endpoint defect. For the prime field, the natural symmetry is the Klein-four action generated by complex conjugation and the functional-equation reflection $s \mapsto 1 - s$; the aperture is the fourfold family $\{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}$; the kernel is Mellin evaluation against the explicit-formula weights; the exhaust channel is the off-seam radial defect; the residue channel is the critical-line oscillation; the scale law is multiplicative dilation in x or additive translation in $u = \log x$; and the target closure rule is the still-unbuilt Q_{prime}.

BBP as a query

The original BBP identity is

$$\pi = \sum_{k=0}^{\infty} \frac{1}{16^k}\left(\frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6}\right),$$

and in the 1997 paper Bailey, Borwein, and Plouffe emphasize both the formula itself and its digit-extraction consequence; Bailey's 2023 compendium repeats the formula and records the later literature on BBP-type constants.

A clean operator-theoretic version is

$$(Q_{\text{BBP}} f) := \sum_{k \geq 0} 16^{-k} \sum_{r=0}^{7} \kappa_r \; f(8k+r), \qquad \kappa = (4,0,0,-2,-1,-1,0,0),$$

so that $Q_{\text{BBP}}(n \mapsto 1/n) = \pi$. Here the operator has exactly the signature structure described above: a residue-wheel symmetry ($\mathbb{Z}/8\mathbb{Z}$), a sparse address aperture ($\{1,4,5,6\}$), a signed mask κ, and a base law 16^{-k}. The discrete signature is not merely combinatorial; it has an exact continuous pullback.

The needed Mellin/Maclaurin derivation is elementary and fully explicit. Start from the geometric expansion

$$\frac{1}{1-x^8} = \sum_{k=0}^{\infty} x^{8k}, \qquad |x| < 1.$$

Then, for $r \in \{1, \dots, 8\}$,

$$\int_0^{1/\sqrt{2}} \frac{x^{r-1}}{1-x^8}\, dx = \sum_{k=0}^{\infty} \int_0^{1/\sqrt{2}} x^{8k+r-1}\, dx = \sum_{k=0}^{\infty} \frac{2^{-r/2}}{16^k(8k+r)}.$$

This is exactly the Mellin-moment relation used in the BBP proof. Multiplying these moments by $4\sqrt{2}$, -8, $-4\sqrt{2}$, and -8 for $r = 1,4,5,6$ respectively gives

$$\pi = \int_0^{1/\sqrt{2}} \frac{4\sqrt{2} - 8x^3 - 4\sqrt{2}x^4 - 8x^5}{1-x^8}\, dx.$$

The BBP series and the integral are therefore equivalent by termwise integration of a Maclaurin expansion.

Now substitute $y = \sqrt{2}\, x$. A short simplification yields

$$\pi = \int_0^1 \frac{16(y-1)}{(y^2-2)(y^2-2y+2)}\, dy = \int_0^1 \left(\frac{4y}{y^2-2} - \frac{4(y-2)}{(y-1)^2+1}\right) dy.$$

This is the exact point at which the BBP "shape" becomes transparent: the first term is logarithmic/radial and the second is angular/arctangent. This decomposition is derived here directly from the published BBP integral, so it is not an analogy or reinterpretation layered on top of the formula.

Define

$$L(y) = 2\log(2 - y^2) - 2\log((y-1)^2 + 1), \qquad \Theta(y) = 4\arctan(y-1).$$

Then

$$L'(y) + \Theta'(y) = \frac{4y}{y^2 - 2} - \frac{4(y-2)}{(y-1)^2 + 1},$$

hence

$$\pi = [L(y) + \Theta(y)]_0^1.$$

Evaluating endpoints gives

$$L(1) - L(0) = 0, \qquad \Theta(1) - \Theta(0) = 0 - 4\arctan(-1) = \pi.$$

So the BBP query annihilates the full radial endpoint defect and reads out exactly one surviving angular residue. That is the mathematically correct form of the project's "radial cancellation / angular survival" statement.

This makes the BBP signature explicit.

Component	Σ_{BBP}
symmetry group	residue-wheel C_8, with discrete dilation by base $16 = 2^4$
address aperture	residues $\{1,4,5,6\} \subset \mathbb{Z}/8\mathbb{Z}$
kernel / mask	$(4,0,0,-2,-1,-1,0,0)$
exhaust channel	logarithmic endpoint terms $L(y)$
residue channel	arctangent phase $\Theta(y)$
base / scale law	$k \mapsto 16^{-k}$
closure rule	$\Delta L = 0$; readout is $\Delta\Theta = \pi$

In the language of the previous section, BBP is a compatible query for a circular/phase invariant because its kernel annihilates the exhaust subspace exactly and leaves an injective residue readout. The prototype is completely rigorous: it follows from the published BBP series and integral representation by straightforward analysis.

Local off-seam detection

For the zeta explicit formula one should begin from the exact fourfold family, not from the seam-only simplification that replaces $1 - \rho$ by $\bar{\rho}$. Let

$$\rho = \frac{1}{2} + \alpha + i\gamma, \qquad u = \log x.$$

Then the four related zeros are $\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}$, and the corresponding contribution to $\psi(x) - x$ is

$$C_\rho(x) = -\left(\frac{x^\rho}{\rho} + \frac{x^{\bar{\rho}}}{\bar{\rho}} + \frac{x^{1-\rho}}{1-\rho} + \frac{x^{1-\bar{\rho}}}{1-\bar{\rho}}\right).$$

Using $x = e^u$, this becomes

$$C_\rho(x) = -2e^{u/2}\Re Z_{\alpha,\gamma}(u),$$

where

$$Z_{\alpha,\gamma}(u) = \frac{e^{(\alpha+i\gamma)u}}{\frac{1}{2}+\alpha+i\gamma} + \frac{e^{-(\alpha+i\ \ )u}}{\frac{1}{2}-\alpha-i\gamma}.$$

This is the correct family carrier. It is exact, and it shows immediately that the off-seam identity $1-\rho=\bar{\rho}$ fails unless $\alpha=0$. That correction is essential.

One can still split $Z_{\alpha,\gamma}$ into even and odd α-parts:

$$ Z_{\alpha,\gamma}(u) = \cosh(\alpha u)\Bigl(\frac{e^{i\gamma u}}{\rho} + \frac{e^{-i\gamma u}}{1-\rho} \Bigr) + \sinh(\alpha u)\Bigl(\frac{e^{i\gamma u}}{\rho} - \frac{e^{-i\gamma u}}{1-\rho} \Bigr). $$

This formula is exact, and it explains precisely what earlier project notes were circling around. The odd $\sinh$-piece is not automatically gone off seam, because its coefficients do not match unless $1-\rho=\bar{\rho}$, i.e. unless $\alpha=0$. Pairing and closure are different things.

The correct local detector is therefore the defect from the seam model:

$$D_{\alpha,\gamma}(u) := Z_{\alpha,\gamma}(u) - Z_{0,\gamma}(u), \qquad Q_{\omega}(\alpha,\gamma) := \int_0^{\infty} \left|D_{\alpha,\gamma}(u)\right|^2 \omega(u)\, du,$$

where $\omega \geq 0$ is a test weight. For the theorem below it is enough to assume

$$\omega \in L^1_{\mathrm{loc}}([0,\infty)), \qquad \omega \geq 0,$$

and that $\omega > 0$ on a set of positive measure on which $D_{\alpha,\gamma}$ is square-integrable. The internal project note already isolates exactly this detector; the proof is short and rigorous.

Theorem. For every such weight ω,

$$Q_{\omega}(\alpha,\gamma) = 0 \Leftrightarrow \alpha = 0.$$

Proof. If $\alpha=0$, then $D_{0,\gamma}\equiv 0$, hence $Q_{\omega}(0,\gamma)=0$. Conversely, suppose $Q_{\omega}(\alpha,\gamma)=0$. Since the integrand is nonnegative, $D_{\alpha,\gamma}(u)=0$ for almost every u in the set where $\omega>0$. That set has an accumulation point because it has positive measure. But $D_{\alpha,\gamma}$ is an entire function of u, being a finite linear combination of exponentials. Hence the identity theorem for real-analytic/entire functions implies $D_{\alpha,\gamma}\equiv 0$ for all u. Expanding both sides,

$$Z_{\alpha,\gamma}(u) = Ae^{(\alpha+i\gamma)u} + Be^{-(\alpha+i\gamma)u}, \qquad Z_{0,\gamma}(u) = Ce^{i\gamma u} + De^{-i\gamma u},$$

with

$$A = \frac{1}{\frac{1}{2}+\alpha+i\gamma}, \quad B = \frac{1}{\frac{1}{2}-\alpha-i\gamma}, \quad C = \frac{1}{\frac{1}{2}+i\gamma}, \quad D = \frac{1}{\frac{1}{2}-i\gamma}.$$

If $\alpha \neq 0$, the exponentials $e^{(\alpha+i\gamma)u}$, $e^{-(\alpha+i\gamma)u}$, $e^{i\gamma u}$, $e^{-i\gamma u}$ have distinct exponents, hence are linearly independent. Equality $Z_{\alpha,\gamma}\equiv Z_{0,\gamma}$ would force $A=0$, impossible. Therefore $\alpha=0$. ■

This theorem is exactly the rigorous local closure statement the project needs. It is also stronger than the earlier "hyperbolic separation energy" based solely on $\cosh(\alpha u)-1$, because $D_{\alpha,\gamma}$ sees the **first-order** off-seam defect coming both from exponentials and from denominators. In fact,

$$D_{\alpha,\gamma}(u) = \alpha\, \partial_{\alpha} Z_{\alpha,\gamma}(u)|_{\alpha=0} + O(\alpha^2),$$

so

$$Q_\omega(\alpha,\gamma) = \alpha^2 c_{\gamma,\omega} + O(\alpha^3), \qquad c_{\gamma,\omega} = \left\|\partial_\alpha Z_{\alpha,\gamma}|_{\alpha=0}\right\|^2_{L^2(\omega)} > 0.$$

Thus the off-seam defect turns on quadratically in the norm, which is exactly the shape expected from a local positive detector.

The corresponding project-status statement is therefore clean:

Statement	Status
exact fourfold family decomposition	rigorous
off-seam simplification $1-\rho=\bar{\rho}$	false unless $\alpha=0$
local detector $Q_\omega(\alpha,\gamma)=0 \Leftrightarrow \alpha=0$	rigorous
PNT by itself excludes off-seam zeros	false
existence of a prime-side global Q_{prime} with no cross-phase escape	open

The last row is the live bolt. It is also where classical RH criteria enter.

Global closure functionals available in the literature

The natural classical home for Q_{prime} is Weil's quadratic functional. In Bombieri's official Clay problem description, Weil's explicit formula is written in Mellin-transform form:

$$\tilde{f}(0) - \sum_\rho \tilde{f}(\rho) + \tilde{f}(1) = \text{prime side},$$

for suitable f, and Bombieri states that RH is equivalent to negativity of the arithmetic side for a special positive class of test functions. Lagarias presents the equivalent positivity version

$$W\left(f * \bar{\tilde{f}}\right) \geq 0 \quad \text{for all nice } f,$$

which is exactly a positive quadratic form criterion. Bombieri's later memoir then shows this quadratic functional is positive semidefinite iff RH and that finite truncations develop negative eigenvalues precisely when off-line zeros exist. This is already extremely close in spirit to the project's goal: a global quadratic form whose failure of positivity measures off-seam defect.

Li's criterion packages the same phenomenon into a discrete sequence. Li proved that RH is equivalent to positivity of

$$\lambda_n = \frac{1}{(n-1)!}\frac{d^n}{ds^n}[s^{n-1}\log\xi(s)]_{s=1},$$

and Bombieri–Lagarias showed both that this can be generalized to arbitrary multisets and that the λ_n are related arithmetically to the Guinand–Weil explicit formula. Lagarias makes the relationship especially transparent by recording that $\lambda_n = W\left(\phi_n * \bar{\tilde{\phi}}_n\right)$. This means Li's coefficients are not a separate world: they are a discrete sampling of Weil's quadratic form. That is mathematically important, but it also explains their limitation for the present project: the basis ϕ_n is global and not naturally localized in (α,γ), so a pullback into diagonal detector terms $\sum c_\rho Q_\omega\left(\alpha_\rho,\gamma_\rho\right)$ is not obvious.

The Nyman–Beurling criterion is the cleanest Hilbert-space closure statement in the literature. Conrey's survey states Nyman's theorem in the $L^2(0,1)$ language of fractional-part functions, and then gives the Balazard–Saias reformulation:

$$\inf_A \int_{-\infty}^{\infty} \left|1 - A\left(\frac{1}{2}+it\right)\zeta\left(\frac{1}{2}+it\right)\right|^2 \frac{dt}{\frac{1}{4}+t^2} = 0,$$

where the infimum is over Dirichlet polynomials A. Báez-Duarte then strengthens the criterion by showing one may restrict to integer dilations. This is the most literal mainstream version of a "closure functional": it is already a positive norm, already lives in Mellin/Fourier space, and already measures whether the prime field closes on the critical line. Its weakness, for the project's purposes, is that localization into individual defect packets $D_{\alpha,\gamma}$ is indirect. The norm is global and highly coupled.

The project's "Gate B weighted Mellin Hilbert-bundle" does not currently exist as a published standard object, but it fits very naturally into known operator-theoretic precedents. Burnol describes an invariant formulation of the explicit formula with a "conductor operator" whose spectral analysis is equivalent to the explicit formula. Connes gives a trace-formula picture in which critical zeros appear as an absorption spectrum and hypothetical noncritical zeros appear as resonances. Deninger, in a different direction, explicitly advocates a cohomological/operator interpretation of analytic number theory. These sources do not by themselves yield Gate B, but they make clear that a Mellin-bundle/operator formulation is mathematically legitimate, not extraneous to the literature.

The candidate landscape can therefore be summarized as follows.

Candidate	Precise classical object	Can it plausibly pull back to local detector terms?	Main obstacle
Weil quadratic form	$W\left(f * \tilde{\bar{f}}\right)$, RH iff positivity	Yes; this is the best classical host for localized Mellin packets	off-diagonal zero coupling and prime-side realization of detector basis
Li coefficients	$\lambda_n \geq 0$ for all n, with $\lambda_n = W\left(\phi_n * \tilde{\bar{\phi}}_n\right)$	Only indirectly; moments are too global	poor localization in (α, γ)
Nyman–Beurling	closure norm $\inf_A \parallel 1 - A\zeta \parallel_{L^2((1/4+t^2)^{-1}dt)}$	Yes, as a closure norm with added seam projector	identifying a detector basis inside the Dirichlet-polynomial closure problem
Gate B Mellin bundle	project-level direct-integral operator built to isolate seam-orthogonal defect	Yes by design	not yet derived from a classical prime-side identity

The ranking is therefore clear. If one wants the closest route to a classical Q_{prime}, one should start from Weil or Nyman–Beurling. Li is best viewed as a derived moment family, not as the primary construction. Gate B is best viewed as a proposed Mellin-bundle **realization** of one of those two classical objects, most naturally of Weil's quadratic form.

Construction program, worked examples, and assumptions

The right construction target is not "show $\cosh(\alpha u) - 1 > 0$, therefore no cancellation." The right target is to build a **prime-side quadratic norm** in which off-seam defect packets form a Riesz sequence. In other words, after whatever localization or bundle transform is chosen, one wants

$$Q_{\text{prime}}\left(\sum_\rho a_\rho D_\rho\right) \geq A \sum_\rho |a_\rho|^2 Q_\omega(\alpha_\rho, \gamma_\rho), \qquad A > 0.$$

This is the exact Hilbert-space meaning of the desired "no-cross-phase-escape theorem." If such a lower frame bound held, then $Q_{\mathrm{prime}} = 0$ would force every $Q_\omega(\alpha_\rho, \gamma_\rho) = 0$, hence every $\alpha_\rho = 0$. Bombieri's eigenvalue results for truncations of the Weil form make this target especially natural.

A concrete Weil-based construction can be written as follows. Choose a smooth compactly supported window $\eta \in C_c^\infty(\mathbb{R})$, and define Mellin packets

$$g_{\alpha,\gamma}(x) = x^{-1/2-\alpha-i} \quad \eta(\log x).$$

Their Mellin transforms are translates of the Fourier transform of η, so they are localized near $s = \frac{1}{2} + \alpha + i\gamma$. If one evaluates the Weil form on $g_{\alpha,\gamma} * \widetilde{\overline{g_{\alpha,\gamma}}}$, then the spectral side naturally samples $\rho \mapsto \hat{g}_{\alpha,\gamma}(\rho)\overline{\hat{g}_{\alpha,\gamma}(1-\bar{\rho})}$. The diagonal hope is that after a seam-orthogonal projection one obtains a form comparable to $Q_\omega(\alpha,\gamma)$. The advantage of compactly supported smooth windows is that off-diagonal inner products become oscillatory integrals with rapid decay in $|\gamma - \gamma'|$. The hard part is proving a uniform lower frame bound over the whole zero set.

A Nyman–Beurling-based construction is even closer to the project's language of closure. In the Mellin variable t, define the residual field

$$\mathcal{R}_A(t) := 1 - A\left(\frac{1}{2} + it\right)\zeta\left(\frac{1}{2} + it\right)$$

in

$$\mathcal{H}_{NB} = L^2\left(\mathbb{R}, \frac{dt}{\frac{1}{4} + t^2}\right).$$

The classical criterion asks whether $\inf_A \parallel \mathcal{R}_A \parallel_{\mathcal{H}_{NB}} = 0$. The project-level refinement would be to insert a seam-orthogonal projector $P_{\perp\mathrm{se}}$ and define

$$Q_{\mathrm{prime}}^{NB}(A) := \parallel P_{\perp\mathrm{seam}} \mathcal{R}_A \parallel_{\mathcal{H}_{NB}}^2.$$

This has the great virtue of already being a bona fide prime-side norm. The missing theorem is again a lower frame bound: one must show that any off-seam zero creates a non-removable seam-orthogonal residue in $\mathcal{R}_A$, uniformly over all Dirichlet-polynomial approximants A.

A formal Gate B realization can be stated cleanly even though it is still conjectural. Let

$$\mathcal{H}_\omega = \int_{\gamma\in\mathbb{R}}^{\oplus} H_{\omega,\gamma}\ d\mu(\gamma), \qquad H_{\omega,\gamma} \cong L^2(\mathbb{R}_u, \omega(u)\, du).$$

Inside each fiber define the seam line

$$S_\gamma := \mathrm{span}\{Z_{0,\gamma}\},$$

and let $P_\perp^\gamma$ be orthogonal projection onto $S_\gamma^\perp$. If a prime-side transform $\mathfrak{T}$ could be built so that the explicit formula becomes

$$\mathfrak{T}(\text{prime data})(\gamma) = \sum_{\rho:\Im\rho=\gamma} a_\rho\ Z_{\alpha_\rho,\gamma},$$

then the natural Gate B closure functional would be

$$Q_B(F) = \int_{\mathbb{R}} \| P_\perp^\gamma F(\gamma) \|_{H_{\omega,\gamma}}^2 \, d\mu(\gamma).$$

Connes's "critical zeros as absorption spectrum / off-line zeros as resonances" picture and Burnol's conductor-operator framework are exactly the operator precedents that make this proposal reasonable. But as of now it remains a formalization, not a theorem.

One instructive worked example is the single-family off-seam packet itself.

Object	Exact formula	Interpretation
family carrier	$Z_{\alpha,\gamma}(u) = \dfrac{e^{(\alpha+i\gamma)u}}{\frac{1}{2}+\alpha+i\gamma} + \dfrac{e^{-(\alpha+i\gamma)u}}{\frac{1}{2}-\alpha-i\gamma}$	exact fourfold family, denominator-correct
seam model	$Z_{0,\gamma}(u) = \dfrac{e^{i\gamma u}}{\frac{1}{2}+i\gamma} + \dfrac{e^{-i\gamma u}}{\frac{1}{2}-i\gamma}$	critical-line oscillation
defect	$D_{\alpha,\gamma} = Z_{\alpha,\gamma} - Z_{0,\gamma}$	local seam-orthogonal residue
detector	$Q_\omega(\alpha,\gamma) = \| D_{\alpha,\gamma} \|_{L^2(\omega)}^2$	positive local closure test
target global theorem	lower frame bound for families of $D_{\alpha,\gamma}$	no-cross-phase escape

A second useful table is the proof roadmap.

Lemma or theorem	Mathematical content	Status
detector closure theorem	$Q_\omega(\alpha,\gamma) = 0 \Leftrightarrow \alpha = 0$	rigorous
packet admissibility	chosen windows lie in Weil or Nyman–Beurling class	mostly straightforward
off-diagonal decay	$\langle D_\rho, D_{\rho'} \rangle_\omega$ decays rapidly in $\lvert\gamma - \gamma'\rvert$ for smooth compact support	rigorous with standard oscillatory-integral estimates
measurable seam projection	$\gamma \mapsto P_\perp^\gamma$ defines a bounded direct-integral projector	straightforward functional analysis
global lower frame bound	defect packets form a Riesz sequence after the chosen prime-side transform	open
prime-side identity	chosen Q_{prime} equals a classical arithmetic functional built from primes	open

The off-diagonal decay row deserves one explicit comment. If $\omega \in C_c^\infty$ and $\rho \neq \rho'$, then

$$\langle D_\rho, D_{\rho'} \rangle_\omega$$

is a finite linear combination of integrals of the form

$$\int \omega(u) e^{(\sigma + i(\gamma - \gamma'))u} \, du.$$

Because ω is smooth and compactly supported, repeated integration by parts gives superpolynomial decay in $|\gamma - \gamma'|$. So a substantial piece of "no cross-phase escape" is not mysterious at all: it is standard almost-orthogonality in the Mellin/Fourier variable. The hard problem is not distant ordinates; it is proving a uniform lower bound for the full infinite Gram matrix.

The most useful visualizations for this program are also clear.

Visualization	What to plot	What it diagnoses
BBP boundary plot	$L(y)$, $\Theta(y)$, and $L'(y)+\Theta'(y)$ on $0 \leq y \leq 1$	exact radial cancellation and angular residue
local defect energy	$\left\|D_{\alpha,\gamma}(u)\right\|^2 \omega(u)$ for $\alpha = 0$ and $\alpha \neq 0$	detector onset and seam orthogonality
radial defect family	$\cosh(\alpha u) - 1$ for several α	why envelope positivity alone is insufficient
Gram heatmap	$G_{\rho,\rho'} = \langle D_\rho, D_{\rho'} \rangle_\omega$ for localized packets	diagonal dominance versus cross-phase leakage
truncation spectrum	eigenvalues of finite Weil-form truncations	Bombieri-style detection of off-line zeros

flowchart LR
L[Local family defect D_{α,γ}] --> Qw[Local detector Q_ω]
Qw --> P[Localized Mellin packets]
P --> W[Weil or NB quadratic norm]
W --> G[Gram matrix / frame bound]
G --> N[No-cross-phase-escape theorem]
N --> Qp[Global closure functional Q_prime]

The assumptions that should be stated explicitly are these. The report uses the standard analytic continuation and functional equation for $\zeta(s)$; nontrivial zeros are counted with multiplicity; the simple explicit formula for $\psi(x)$ is interpreted in the conventional truncated sense and away from prime powers; Mellin-transform conventions differ by a shift across sources, and this report uses Bombieri's $x^s\,dx/x$ convention in the Weil section while retaining the usual $x^{s-1}dx$ form in elementary BBP manipulations; detector weights ω are assumed nonnegative and positive on a set of positive measure, with C_c^∞ convenient for localization arguments; and, most importantly, no argument in the report assumes the RH-equivalent square-root error term in order to prove RH. The open problem is to derive a new positive prime-side form, not to smuggle in an equivalent estimate.

The final conclusion is therefore sharp. The correct rigorous formalization of the project's "Universal Action Type System" is a seven-component signature calculus of symmetry, aperture, mask, channel split, scale law, and positive closure rule. BBP is the exact prototype where the calculus can be carried through to the end. The single-family detector Q_ω is rigorous and useful. The global RH frontier is the construction of a prime-side quadratic norm with a lower frame bound for seam-orthogonal defect packets. In the classical literature, the most promising hosts are Weil's quadratic form and the Nyman–Beurling closure norm; Li's coefficients are best treated as a moment family derived from Weil; and a Gate B weighted Mellin bundle is best understood as a conjectural operator realization modeled on Burnol and Connes. That is the mathematically sound expansion path.

Universal Action Type System and Prime-Field Closure Extented

Executive summary

This report formalizes the user's "Universal Action Type System" as a mathematical schema for **queries that become invariant read-heads at a seam**: one channel exhausts, another survives, and closure is the condition that makes the surviving residue globally stable. The scheme is then applied to the Riemann explicit formula to isolate a rigorous, single-family **radial-separation detector** $Q_\omega(\alpha, \gamma)$, which vanishes exactly on the critical seam $\alpha = 0$ under mild hypotheses on the weight ω. That part is mathematically straightforward. The hard part is not local detection but global closure: building a host functional Q_{prime} that forces these local defects to survive summation over all zero families without being neutralized by cross-terms. That is exactly where the current literature still leaves the Riemann Hypothesis open. The most promising host functional is André Weil's quadratic form, because Weil's criterion is already an RH-equivalent positivity statement built directly from the explicit formula. Li's coefficients are closely related to Weil's form but are less naturally localized by individual zero families. The Nyman–Beurling distance is the strongest established "closure problem" formulation, but it is naturally global and Hardy-space-based rather than family-diagonal. A recent public "Gate B" note suggests a weighted Mellin-space operator framework, but in the publicly available form it is still too underspecified for a publishable proof engine. RH itself remains open.

The highest-confidence mathematical output of this report is therefore not a proof of RH but a clean separation of what is already formal from what must still be built:

$$\boxed{\text{Single-family defect detection is easy; global prime-field closure is the real bolt.}}$$

More concretely, for the exact denominator-aware zero-family model

$$Z_{\alpha,\gamma}(u) = \frac{e^{(\alpha+i\gamma)u}}{\frac{1}{2} + \alpha + i\gamma} + \frac{e^{-(\alpha+i\gamma)u}}{\frac{1}{2} - \alpha - i\gamma}, \qquad D_{\alpha,\gamma}(u) = Z_{\alpha,\gamma}(u) - Z_{0,\gamma}(u),$$

the separation energy

$$Q_\omega(\alpha, \gamma) = \int_I \left|D_{\alpha,\gamma}(u)\right|^2 \omega(u)\, du$$

is nonnegative and, under minimal support assumptions on ω, satisfies

$$Q_\omega(\alpha, \gamma) = 0 \Leftrightarrow \alpha = 0.$$

This is a legitimate “Separation Energy Lemma.” What is not yet in hand is a global functional Q_{prime} for which one can prove a lower bound of the schematic form

$$Q_{\mathrm{prime}} \gtrsim \sum_{\rho} c_\rho \; Q_{\omega_\rho}(\beta_\rho - 1/2\,, \gamma_\rho), \qquad c_\rho > 0,$$

or else prove that no such diagonalization is possible in a sufficiently broad class. That is the publishable Section 3I target.

The strongest recommendation is to pursue **localized Mellin packets inside Weil’s quadratic form** as the main line, with **Nyman–Beurling / Báez-Duarte** as the backup line. Li’s criterion should be treated as a derived coordinate system on Weil rather than as the primary host. The “Gate B weighted Mellin bundle” is worth treating as an experimental operator model, but not yet as a proof substrate unless its operator class, measure, functional-equation normalization, and prime-to-spectrum dictionary are written down in a conventionally verifiable way.

Universal Action Type System

I will use **action** and **query** interchangeably. A Universal Action Type System is not a new theorem from the literature; it is a proposed formal language for organizing structures that already appear in BBP, the Riemann explicit formula, and SHA-256. The core datum is the **parity signature**

$$\Sigma(Q) = \big(S(Q), A(Q), K(Q), E(Q), R(Q), B(Q), C(Q)\big),$$

attached to a query Q.

The components are defined as follows.

Let Q act on a state space X_Q and be analyzed in a function/operator space $\mathcal{H}_Q$.

$$\Sigma(Q) = (S, A, K, E, R, B, C)$$

with the meanings:

Component	Definition
$S(Q)$ symmetry	The involution, conjugation, reflection, or finite group action that pairs admissible states. Typically $J^2 = \mathrm{Id}$.
$A(Q)$ aperture	The admissible window, support, residue classes, word-width, or spectral band on which Q actually probes the system.
$K(Q)$ kernel	The exactness set of the read-head: parameters where the defect or detector vanishes.
$E(Q)$ exhaust	The channel compelled to vanish at closure or seam.
$R(Q)$ residue	The complementary invariant that survives after exhaust.
$B(Q)$ base	The intrinsic scale of the query: digit base, word-width, Mellin base, or modular period.
$C(Q)$ closure	The global condition under which the residue becomes a stable invariant and the query is a true read-head rather than a noisy probe.

In this language, a **seam** is any subset of parameters on which the exhaust is exact and closure holds. A **BBP-style invariant read-head** is then simply a query for which $E(Q)$ vanishes on the seam while $R(Q)$ remains nontrivial and stable.

Three formal examples fit naturally into this schema.

The first is the Bailey–Borwein–Plouffe hexadecimal formula for π,

$$\pi = \sum_{i=0}^{\infty} \frac{1}{16^i}\left(\frac{4}{8i+1} - \frac{2}{8i+4} - \frac{1}{8i+5} - \frac{1}{8i+6}\right),$$

together with its integral form

$$\pi = \int_0^{1/\sqrt{2}} \frac{4\sqrt{2} - 8x^3 - 4\sqrt{2}x^4 - 8x^5}{1 - x^8}\, dx.$$

These are the original BBP identities. For the present type-system formalization, one fixes a primitive $Q_{\mathrm{BBP}}(t) = L(t) + i\Theta(t)$ after partial fractions, normalized so that the angular part may be taken as $\Theta(t) = 4\arctan(\sqrt{2}\,t - 1)$, while $L(t)$ is a real logarithmic combination. In that normalization the BBP seam occurs at the endpoint pair $t = 0$ and $t = 1/\sqrt{2} = 16^{-1/8}$: the logarithmic increment cancels,

$$\Delta L = 0,$$

while the angular increment survives,

$$\Delta\Theta = \pi.$$

The published BBP paper supplies the base-16 / mod-8 structure and the integral identity; the $L + \Theta$ splitting used here is a formal normalization choice for the UATS example.

The second example is the Riemann explicit formula. In Conrey's normalization, for suitable f,

$$T(f) := \sum_{\rho} \tilde{f}\,(\rho)$$

equals a prime-and-archimedean expression, and RH is equivalent to positivity on multiplicative self-convolutions $f * f^*$. Weil's original 1952 paper states the same principle in a more general adelic language: RH is equivalent to nonnegativity of the explicit-formula distribution on functions of the form $F_0 * F_0^-$. This makes the explicit-formula zero family a canonical UATS object: the symmetry is the quartet $\{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}$, the seam is the critical line $\Re\rho = 1/2$, the exhaust is the odd hyperbolic part induced by pairing, and the residue is the even radial defect that disappears only on the seam.

The third example is SHA-256. The official object here is not a "trace projector" but the collection of 32-bit bitwise maps in FIPS 180-4, especially

$$\Sigma_0^{\{256\}}(x) = \mathrm{ROTR}^2(x) \oplus \mathrm{ROTR}^{13}(x) \oplus \mathrm{ROTR}^{22}(x),$$

$$\Sigma_1^{\{256\}}(x) = \mathrm{ROTR}^6(x) \oplus \mathrm{ROTR}^{11}(x) \oplus \mathrm{ROTR}^{25}(x).$$

The standard gives the rotations; the **trace projector**

$$P_{\mathrm{tr}}(w) := \Sigma_0(w) \oplus \Sigma_1(w)$$

is an experiment-specific abstraction, not a NIST primitive. In the motivating experiments, the seam word is taken to be a specific 32-bit kernel point; in your notes that point is $0x\mathrm{FFFFFFFF}$, but that claim is not part of the SHA-256 standard and should be labeled empirical unless a separate proof is supplied. What is official and stable is the 32-bit rotational symmetry built into Σ_0, Σ_1.

The three signatures can therefore be organized as follows.

Example	$S(Q)$ symmetry	$A(Q)$ aperture	$K(Q)$ kernel	$E(Q)$ exhaust	$R(Q)$ residue	$B(Q)$ base	$C(Q)$ closure
BBP read-head	octic / mod-8 residue symmetry	residues $\{1,4,5,6\}$, $t \in [0, 2^{-1/2}]$	endpoint pair $\{0, 2^{-1/2}\}$	$\Delta L = 0$	$\Delta\Theta = \pi$	$16 = 2^4$	$2^{-1/2} = 16^{-1/8}$ and endpoint exactness
explicit-formula family	$\rho \leftrightarrow \bar{\rho}$, $\rho \leftrightarrow 1-\rho$	log-scale window $u \in I$, weight ω	$\alpha = 0$ for $D_{\alpha,\gamma}$	odd sinh-part under pairing	even cosh-defect	Mellin base $x = e^u$	$1-\rho = \bar{\rho}$ iff $\alpha = 0$
SHA trace projector	cyclic bit-rotation symmetry	32-bit words	$P_{\text{tr}}(w) = 0$	rotation-gap collapse at seam	carry / residual phase	2^{32} word arithmetic	experiment-specific seam condition

This is the cleanest formal way to say what your notes repeatedly observe: the same pattern can recur across domains without pretending that the domains are literally identical.

Detector and separation energy

The correct zero-family contribution must respect the different denominators of ρ and $1-\rho$. Let

$$\rho = \frac{1}{2} + \alpha + i\gamma, \qquad u = \log x.$$

The contribution of the fourfold family

$$\{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}$$

to the classical explicit formula for $\psi(x)$ is

$$C_\rho(x) = -\left(\frac{x^\rho}{\rho} + \frac{x^{\bar{\rho}}}{\bar{\rho}} + \frac{x^{1-\rho}}{1-\rho} + \frac{x^{1-\bar{\rho}}}{1-\bar{\rho}}\right).$$

Grouping conjugates gives

$$C_\rho(x) = -2e^{u/2}\operatorname{Re}Z_{\alpha,\gamma}(u),$$

where

$$Z_{\alpha,\gamma}(u) := \frac{e^{(\alpha+i\gamma)u}}{\frac{1}{2} + \alpha + i\gamma} + \frac{e^{-(\alpha+i\gamma)u}}{\frac{1}{2} - \alpha - i\gamma}.$$

The seam-subtracted detector is

$$D_{\alpha,\gamma}(u) := Z_{\alpha,\gamma}(u) - Z_{0,\gamma}(u).$$

When $\alpha = 0$, one has $1 - \rho = \bar{\rho}$, so the mirror denominators align exactly and $D_{0,\gamma} \equiv 0$. This is the precise algebraic meaning of "denominator closure on the seam." The explicit formula and Weil positivity criterion in this multiplicative/Mellin normalization are standard.

A useful split is

$$Z_{\alpha,\gamma}(u) = \cosh(\alpha u)\, A_{\alpha,\gamma}(u) + \sinh(\alpha u)\, B_{\alpha,\gamma}(u),$$

with

$$A_{\alpha,\gamma}(u) := \frac{e^{i\gamma u}}{\frac{1}{2} + \alpha + i\gamma} + \frac{e^{-i\gamma u}}{\frac{1}{2} - \alpha - i\gamma},$$

$$B_{\alpha,\gamma}(u) := \frac{e^{i\gamma u}}{\frac{1}{2} + \alpha + i\gamma} - \frac{e^{-i\gamma u}}{\frac{1}{2} - \alpha - i\gamma}.$$

Under the functional-equation pairing $\alpha \leftrightarrow -\alpha$, the $\sinh$-part is the odd channel and the $\cosh$-part is the even channel. This is the exact, denominator-aware version of the heuristic "odd pressure cancels, even pressure persists."

Now fix an interval $I \subseteq \mathbb{R}$ or $I = (0, \infty)$ and define, for a nonnegative measurable weight ω,

$$Q_\omega(\alpha, \gamma) := \int_I \left|D_{\alpha,\gamma}(u)\right|^2 \omega(u)\, du.$$

The main deterministic statement is then the following.

Separation Energy Lemma

Assume:

1. $\omega \geq 0$ is measurable and not identically zero.
2. The set $\{u \in I : \omega(u) > 0\}$ has an accumulation point in I. An open interval on which $\omega > 0$ a.e. is more than enough.
3. $D_{\alpha,\gamma} \in L^2(I, \omega(u)\, du)$ for the parameter range under study.

Then for every admissible (α, γ),

$$Q_\omega(\alpha, \gamma) \geq 0,$$

and

$$Q_\omega(\alpha, \gamma) = 0 \Leftrightarrow \alpha = 0.$$

Proof

Nonnegativity is immediate from the definition.

If $\alpha = 0$, then $D_{0,\gamma} \equiv 0$, so $Q_\omega(0, \gamma) = 0$.

Conversely, if $Q_\omega(\alpha, \gamma) = 0$, then $D_{\alpha,\gamma}(u) = 0$ for ω-almost every u in the positive-support set of ω. But $D_{\alpha,\gamma}(u)$ is an entire function of u, being a finite linear combination of exponentials. An entire function that vanishes on a set with an accumulation point vanishes identically. Hence

$$D_{\alpha,\gamma}(u) \equiv 0.$$

If $\alpha \neq 0$, then $D_{\alpha,\gamma}$ is a nontrivial linear combination of exponentials with distinct exponents:

$$e^{(\alpha+i\gamma)u}, \quad e^{-(\alpha+i\gamma)u}, \quad e^{i\gamma u}, \quad e^{-i\gamma u},$$

except in the degenerate case $\gamma = 0$, where the set reduces to

$$e^{\alpha u}, \quad e^{-\alpha u}, \quad 1,$$

which is still linearly independent when $\alpha \neq 0$. Therefore $D_{\alpha,\gamma} \equiv 0$ is impossible unless $\alpha = 0$. This proves the equivalence.

So the detector really does separate off-seam families from seam families:

$$\boxed{Q_\omega(\alpha,\gamma) = 0 \Leftrightarrow \alpha = 0.}$$

This is the mathematically correct form of the "hyperbolic tear" observation. It is rigorous, but only local in zero-family space.

Analyticity and minimal conditions on ω

For each fixed u, both $Z_{\alpha,\gamma}(u)$ and $D_{\alpha,\gamma}(u)$ are real-analytic in (α,γ) on the strip

$$-1/2 < \alpha < 1/2, \qquad \gamma \in \mathbb{R},$$

because the denominators stay nonzero there and the exponentials are entire.

For the map $(\alpha,\gamma) \mapsto Q_\omega(\alpha,\gamma)$, a sufficient condition for real-analyticity on any compact $|\alpha| \leq A < 1/2$ is that differentiation under the integral be legal. The cleanest sufficient hypotheses are:

- **compactly supported weight** ω, or more generally
- for every $m \geq 0$,

$$\int_I (1+|u|)^m \, e^{2Au} \, \omega(u) \, du < \infty \quad \text{if } I = (0,\infty),$$

  and the corresponding two-sided exponential moment condition if $I = \mathbb{R}$.

Under these hypotheses one differentiates $D_{\alpha,\gamma}$ termwise and applies dominated convergence. For the mere equivalence

$$Q_\omega = 0 \Leftrightarrow \alpha = 0,$$

compact support is **not** needed; the earlier support-plus-integrability assumptions are enough.

The subtle but important conclusion is this:

The detector is already a complete local seam test. What it is not, by itself, is a global prime-field closure theorem.

That distinction is exactly what prevents the "single-family domination" argument from already being a proof of RH.

Candidate global closure functionals

This section asks for a host functional

$$Q_{\mathrm{prime}}$$

capable, in some precise sense, of turning the local detector energy $Q_\omega(\alpha,\gamma)$ into a global obstruction to off-seam zeros. The literature offers four serious candidates.

Weil quadratic form

This is the most natural host because it is already an RH-equivalent positivity criterion built from the explicit formula.

In Conrey's normalization, for $f \in C_0^\infty(0,\infty)$,

$$T(f)=\sum_{\rho}\tilde{f}(\rho)$$

equals

$$\int_0^\infty f(x)\,dx+\int_0^\infty f^*(x)\,dx-\sum_{n=1}^{\infty}\Lambda(n)\big(f(n)+f^*(n)\big)-(\log 4\pi+\gamma)f(1)-\int_1^\infty\left(f(x)-f^*(x)-\frac{2}{x}f(1)\right)\frac{x\,dx}{x^2-1},$$

where

$$f^*(x)=\frac{1}{x}f(1/x),\qquad \tilde{f}(s)=\int_0^\infty f(x)x^{s-1}\,dx.$$

He then states the RH-equivalent positivity condition

$$\sum_{\rho}\tilde{g}(\rho)\,\tilde{g}^*(1-\rho)>0$$

for every nonzero smooth compactly supported g. Weil's original 1952 paper states the same criterion in the test-function class $F=F_0*F_0^-$: RH holds iff the common value of the explicit formula is nonnegative for all such F. Burnol later emphasized the version

$$\sum_{\rho}\hat{g}(\rho)\hat{g}(1-\rho)\geq 0$$

for all smooth compactly supported multiplicative test functions g. Bombieri's work treats this as a genuine quadratic functional, proves that support-restricted minimizers exist, and studies its small-support positivity.

A clean formal definition is therefore:

$$Q_{\text{Weil}}(g):=T(g*g^\tau),\qquad g^\tau(x)=x^{-1}\overline{g(x^{-1})}.$$

This is positive semidefinite for all admissible g if and only if RH holds.

The attraction of Q_{Weil} is obvious: it is already the right global closure functional in principle. The problem is diagonalization. For a packet decomposition

$$g=\sum_j c_j\,g_j,$$

one gets

$$Q_{\text{Weil}}(g)=\sum_{i,j}c_i\,\overline{c_j}\,T\big(g_i*g_j^\tau\big),$$

so the off-diagonal terms are built in from the start. To recover something like

$$Q_{\text{Weil}}(g)\gtrsim\sum_j c_j\;Q_{\omega_j}\big(\alpha_j,\gamma_j\big),$$

one needs a basis or frame $\{g_j\}$ for which the Gram matrix is asymptotically diagonal. That is a real analytical problem, not a formal one.

The main technical obstacles are:

- cross-family interference $T\left(g_i * g_j^{\tau}\right)$ for $i \neq j$,
- archimedean and pole terms that may not localize familywise,
- denominator geometry: the detector $D_{\alpha,\gamma}$ lives in the exact $\rho/1-\rho$ geometry, while Q_{Weil} sees only Mellin transforms of test functions,
- proving almost-orthogonality without smuggling RH into the argument.

Even with those obstacles, this is the strongest candidate.

Li coefficients

Li's criterion is another RH-equivalent positivity system, but it packages the explicit formula differently. Li proved that RH for the Dedekind zeta function is equivalent to the nonnegativity of a sequence of real numbers. For the Riemann zeta function, in the standard notation,

$$\lambda_n = \sum_{\rho} \left(1 - \left(1 - \frac{1}{\rho}\right)^n\right) = \frac{1}{(n-1)!} \frac{d^n}{ds^n} \left(s^{n-1} \log \xi(s)\right)\Big|_{s=1},$$

and RH is equivalent to

$$\lambda_n \geq 0 \qquad (n \geq 1).$$

Conrey also notes that the Li test functions g_n can be interpreted as a special sequence inside Weil's positivity framework, while Lagarias proved in the automorphic setting that generalized Li coefficients are related to values of Weil's quadratic functional on a suitable test-function space.

A precise functional candidate is therefore any positive combination

$$Q_{\mathrm{Li}}(a) := \sum_{n \geq 1} a_n \lambda_n$$

or quadratic refinement thereof.

The advantage is that λ_n is already a sum over zeros, so it looks "diagonal" with respect to zero index. But this is deceptive. The individual per-zero summands

$$1 - \left(1 - \frac{1}{\rho}\right)^n$$

are generally complex and highly oscillatory as functions of n. Positivity only emerges after summing over the whole zero set and, on the seam, over mirror pairs. So Li coefficients do not natively produce a positive defect of the form

$$(\beta - 1/2)^2 w(\gamma)$$

for each family.

The principal obstacles are:

- the summands are not positive familywise,
- cross-n interactions are hidden in the generating function rather than eliminated,
- the geometry is rational in $1/\rho$, not local in the log-scale variable u,
- any quadratic packaging that restores positivity tends to collapse back toward Weil's form.

Thus Li is valuable, but as a coordinate system on Weil rather than as the cleanest localization platform.

Nyman–Beurling distance

This is the most famous RH-as-closure formulation, and in some ways the conceptually closest to the phrase "prime-field closure functional."

Balazard and Saias explain the Mellin-space reformulation clearly. If $\rho(t)$ denotes the fractional-part kernel, then RH is equivalent to the density statement

$$\overline{\mathrm{span}}_{L^2(0,1)}\{\rho_\alpha\colon\ 0<\alpha<1\}=L^2(0,1),$$

and a sharpened formulation says RH is already equivalent to the characteristic function $\chi_{(0,1)}$ lying in the closure of this span. Under the Mellin transform, Bercovici–Foias identify the closure in terms of holomorphy of $Mf(s)/\zeta(s)$ in the half-plane $\Re s>1/2$. Báez-Duarte's strengthening restricts the dilation parameters to integers. Balazard–Saias also record the finite-dimensional approximation problem

$$d_n=\mathrm{dist}_{L^2(0,\infty)}\left(\chi,\sum_{k=1}^{n}a_k\ \rho\big(1/(kt)\big)\right),$$

and a lower bound of order $1/\sqrt{\log n}$ involving the zeros on the critical line.

A precise candidate functional is therefore

$$Q_{\mathrm{NB}}(N):=\inf_{a_1,\dots,a_N}\left\|\chi-\sum_{n\le N}a_n\,\rho_{1/n}\right\|_{L^2}^2.$$

This is unquestionably a closure functional. The issue is diagonalization. In Mellin space, the obstruction created by zeros of ζ is encoded through inner/Hardy factorization of ζ, not through a sum of isolated positive family defects. Off-seam zeros are global Hardy-space obstructions, not automatically orthogonalizable detector atoms.

The major obstacles are:

- the natural language is invariant-subspace / Hardy-space closure, not familywise positivity,
- zeros appear via divisibility and factorization constraints on Mf/ζ, not through a pre-diagonal quadratic energy,
- finite-dimensional approximants involve Dirichlet polynomials, so frequency mixing is structural,
- the lower bounds known in the literature do not yet isolate $(\beta-1/2)^2$-type contributions family by family.

Still, this route is extremely attractive because "closure" is not metaphorical here; it is literally the theorem statement.

Gate B weighted Mellin Hilbert bundle

A recent public Zenodo preprint describes "Gate B" as a two-fiber weighted Mellin-space Hilbert bundle with a mirror operator implementing the functional equation, a closed-loop operator, and quasi-compactness / Hennion-style claims for a renormalized transfer operator. In the publicly available text, the claimed space is a weighted Mellin-space Hilbert bundle built from a sub-invariant measure, the mirror is a two-fiber involution, and the closure problem is rephrased as invertibility of a Schur complement or contraction of a closed-loop operator off the seam.

As a **published mathematical definition**, however, the current public note is still incomplete for the present purpose. The following items are not yet specified at the level needed for a conventional proof:

- the exact operator T representing "arithmetic runtime,"

- the exact measure ω or μ on the Mellin/log side,
- the precise relation between the operator spectrum and the zeros of ζ,
- the exact normalization of the mirror factor reproducing $s \mapsto 1 - s$,
- proofs that the claimed boundedness, quasi-compactness, and determinant-class properties hold in the stated space.

So the mathematically honest status is this: Gate B is a **plausible design pattern**, not yet an established closure functional in the literature.

A serviceable formal surrogate would be

$$\mathcal{H}_\omega = L^2(\mathbb{R}, \omega(u)\, du) \oplus L^2(\mathbb{R}, \omega(u)\, du),$$

with a mirror operator J implementing $u \mapsto -u$ and seam/unitarity normalization, an arithmetic transfer T, and a closed-loop functional

$$Q_{\mathrm{GateB}}(f) = \langle (I - L^* L) f, f \rangle, \qquad L = TJTJ.$$

But that is a proposal, not yet a sourced theorem.

Comparative assessment

On present evidence, the four candidates line up as follows.

Candidate	Precise status	Natural positivity	Natural closure	Family-diagonal prospects	Main obstacle
Weil quadratic form	established, RH-equivalent	very strong	strong	moderate	cross-packet localization
Li coefficients	established, RH-equivalent	strong but indirect	moderate	low-to-moderate	oscillatory per-zero contributions
Nyman–Beurling distance	established, RH-equivalent	moderate	very strong	low	Hardy/global rather than diagonal
Gate B weighted Mellin bundle	speculative / public-preprint level	unknown	intended to be strong	unknown	underspecified operator theory

If the goal is a publishable Section 3I, the order of attack should be: **Weil first, Nyman–Beurling second, Li as an auxiliary coordinate system, Gate B as an experimental operator template.**

Construction strategies and proof architecture

The goal of this section is not to claim a finished Q_{prime}, but to state the most concrete routes to one.

Localized Mellin packets inside Weil's form

This is the best path.

Choose a weighted log-scale Hilbert space

$$H_\omega := L^2(I, \omega(u)\, du),$$

with $I = (0, \infty)$ or a compact subinterval, and define the seam packet family at frequency γ by

$$s_\gamma(u) := \frac{e^{i\gamma u}}{\frac{1}{2} + i\gamma} + \frac{e^{-i\gamma u}}{\frac{1}{2} - i\gamma},$$

and the off-seam family packet by

$$z_{\alpha,\gamma}(u) := \frac{e^{(\alpha+i\gamma)u}}{\frac{1}{2} + \alpha + i\gamma} + \frac{e^{-(\alpha+i\gamma)u}}{\frac{1}{2} - \alpha - i\gamma}.$$

The detector is then

$$D_{\alpha,\gamma} = z_{\alpha,\gamma} - s_\gamma.$$

Define the **seam projection**

$$P_{\perp\,\mathrm{seam}}: H_\omega \to H_\omega$$

as the orthogonal projection onto the complement of the span of seam packets (or onto the complement of a finite/packetized approximation to that span). Then

$$Q_\omega(\alpha,\gamma) = \| P_{\perp\,\mathrm{seam}} z_{\alpha,\gamma} \|_{H_\omega}^2.$$

The problem is to realize these packets as Mellin images of actual Weil test functions g, and to prove that the restricted Weil Gram matrix is asymptotically diagonal. The following lemmas would make the program work.

Lemma A. Mellin packet realization. For each detector packet $D_{\alpha,\gamma}$ in a chosen compact (α,γ)-window, there exists a smooth compactly supported multiplicative test function $g_{\alpha,\gamma}$ such that $\tilde{g}_{\alpha,\gamma}$ localizes near (α,γ) and

$$\tilde{g}_{\alpha,\gamma}(\rho) \approx D_{\alpha,\gamma}$$

in the desired packet norm.

Lemma B. Almost-orthogonality. For packet families with sufficiently separated γ-parameters,

$$T\left(g_i * g_j^\tau\right) = o(1) \qquad (i \neq j)$$

as the support scale or localization parameter increases, while

$$T(g_i * g_i^\tau) \asymp Q_{\omega_i}(\alpha_i, \gamma_i).$$

Lemma C. Archimedean remainder bound. The prime/pole/archimedean side of Weil's formula differs from the packet-diagonal model by an error bounded by

$$\varepsilon \sum_i |c_i|^2$$

for the same localization regime.

If these three lemmas hold, then one obtains

$$Q_{\mathrm{Weil}}(g) \geq \sum_i c_i\ Q_{\omega_i}(\alpha_i, \gamma_i) - \varepsilon \sum_i |c_i|^2,$$

which is exactly the kind of closure inequality Section 3I needs.

The natural proof tools are Mellin/Fourier packet localization, Plancherel, stationary phase on the log line, and Gram-matrix estimates. Burnol's convolution-stable class and Weil's smooth compactly supported test functions are the right ambient spaces. Hennion-style quasi-compactness ideas become relevant only if one upgrades the packet story into an actual transfer operator.

Li-generated positive kernels

A second approach is to use **positive combinations of Li coefficients** to synthesize a better zero kernel. Since Lagarias shows the generalized Li coefficients are tied to values of Weil's quadratic functional, one can try to choose coefficients a_n so that

$$\sum_{n\geq 1} a_n \lambda_n$$

behaves like a positive transform of

$$(\beta - 1/2)^2$$

after pairing the zero family. In practice, this means finding a generating kernel $K(\rho)$ that is positive on off-seam pairs and vanishes on seam pairs. The difficulty is that the primitive Li summands

$$1 - \left(1 - \frac{1}{\rho}\right)^n$$

are not individually positive, so positivity must come from a Herglotz-type or square-kernel construction rather than a naive positive coefficient choice. This is plausible, but it almost certainly folds back into a disguised Weil form.

Nyman–Beurling seam projection

The third route is to move closure to where closure is already the theorem statement.

On the Mellin side, the Nyman–Beurling criterion says that RH is equivalent to density of a dilation span, or equivalently to the closure of χ inside the corresponding span. The obvious experimental construction is to define a seam-projected distance

$$Q_{\mathrm{NB},\perp} := \inf_{f \in \overline{\mathrm{span}}(\rho_{1/n})} \| P_{\perp\,\mathrm{seam}}(\chi - f) \|_{L^2}^2$$

and try to prove that off-seam zeros force a positive lower bound that decomposes into detector contributions.

The right tools would be Hardy-space factorization, Blaschke products, Carleson interpolation, and Báez-Duarte-type optimization over Dirichlet polynomials. The main difficulty is conceptual: Nyman–Beurling encodes zeros through invariant-subspace structure, not through an already-isolated positive energy per zero family. So one may get a closure theorem first and family-diagonalization second, rather than the other way around.

A mathematically serviceable Gate B model

If one wants to preserve the two-fiber mirror language, the cleanest rigorous version would be to begin with a **completely abstract** doubled Hilbert space

$$\mathcal{H}_\omega = L^2(\mathbb{R}, \omega(u)\, du) \oplus L^2(\mathbb{R}, \omega(u)\, du),$$

a unitary seam mirror

$$J_0(f_+, f_-)(u) = \big(f_-(-u), f_+(-u)\big),$$

an off-seam weighted mirror

$$J_\alpha(f_+, f_-)(u) = \left(e^{\alpha u} f_-(-u), e^{-\alpha}\ \ f_+(-u)\right),$$

and a bounded positive arithmetic operator T commuting suitably with the seam mirror. Then the closed-loop operator

$$L_\alpha = T J_\alpha T J_\alpha$$

suggests the candidate closure functional

$$Q_{\mathrm{GateB}}(f) := \langle (I - L_\alpha^* L_\alpha) f, f \rangle.$$

The right theorem would be

$$I - L_\alpha^* L_\alpha \geq \kappa(\alpha)\, P_{\perp\,\mathrm{seam}}, \qquad \kappa(\alpha) > 0 \text{ for } \alpha \neq 0,$$

with $\kappa(0) = 0$. If proved in a model genuinely attached to the explicit formula or prime transfer, this would be a true closure functional.

At present, that is a design specification, not a theorem. The public Gate B note has the correct kind of ingredients — weighted Mellin space, mirror bundle, quasi-compactness rhetoric, Schur complement language — but not yet in a form that can be independently checked line by line as a proof of RH.

Dependency graph

flowchart TD

A[Exact zero-family packet Z_{α,γ}] --> B[Detector D_{α,γ}=Z_{α,γ}-Z_{0,γ}]
B --> C[Single-family lemma Q_ω(α,γ)=0 iff α=0]
C --> D[Choose host space H_ω and seam projection P_perp]
D --> E[Build localized Mellin packets]
E --> F[Prove Gram almost-orthogonality]
F --> G[Embed packets into host functional]
G --> H{Host functional}
H --> I[Weil quadratic form]
H --> J[Li-generated kernel]
H --> K[Nyman-Beurling distance]
H --> L[Gate B operator model]
I --> M[Diagonal lower bound or obstruction]
J --> M
K --> M
L --> M
M --> N[Section 3I: Q_prime constructed or impossibility theorem]

Toy models and numerical illustrations

The following experiments are not claims from the literature; they are straightforward numerical illustrations of the detector formalism.

For all experiments below, I took the compactly supported weight

$$\omega(u) = \mathbf{1}_{[0,8]}(u),$$

so

$$Q_{[0,8]}(\alpha,\gamma) = \int_0^8 \left|D_{\alpha,\gamma}(u)\right|^2 \, du.$$

Single-family scaling

Using γ values near the first zeta zeros, one gets:

α	γ	$Q_{[0,8]}(\alpha,\gamma)$
0.00	14.1347	0.000000
0.01	14.1347	0.000171
0.05	14.1347	0.004509
0.10	14.1347	0.021148
0.10	21.0220	0.009546
0.20	14.1347	0.148415

The behavior is exactly what the theorem predicts: the detector is zero on the seam and grows smoothly once $\alpha \neq 0$. The dependence on γ is nontrivial because the denominators matter, but positivity persists.

Distinct packets do not cancel completely

For the three synthetic off-seam families

$$(0.10, 14.1347), \qquad (0.08, 21.0220), \qquad (0.06, 25.0109),$$

the Gram matrix

$$G_{ij} = \int_0^8 \overline{D_i(u)}\, D_j(u)\, du$$

has eigenvalues approximately

$$0.002094, \qquad 0.005642, \qquad 0.021188.$$

So G is positive definite. The normalized overlaps are small:

$$|G_{12}|/\sqrt{G_{11}G_{22}} \approx 0.069, \qquad |G_{13}|/\sqrt{G_{11}G_{33}} \approx 0.039, \qquad |G_{23}|/\sqrt{G_{22}G_{33}} \approx 0.065.$$

This is the ideal regime for a successful diagonalization theorem: separated packets, small cross-terms, positive Gram spectrum.

Near-duplicate packets can almost cancel

For

$$(0.10, 14.1347), \qquad (0.12, 14.1347),$$

the normalized overlap is about

$$0.998653,$$

and the smallest Gram eigenvalue is only about

$$3.48 \times 10^{-5}.$$

With coefficients $(1, -1)$, the energy drops to about

$$0.00144.$$

So cross-term cancellation can be **very strong** when packets overlap heavily. This is not a bug; it is exactly why localization and orthogonalization matter.

Exact duplicates cancel totally

If one includes the same detector twice, namely

$$(0.10, 14.1347), \qquad (0.10, 14.1347),$$

the Gram matrix has eigenvalues

$$0, \qquad 0.042295.$$

The coefficient vector $(1, -1)$ annihilates the energy exactly.

This trivial model makes the proof obligation perfectly clear:

$$\boxed{\text{A global closure theorem needs either genuine orthogonality, or a quotient that removes duplicate/near-duplicate packet directions.}}$$

That is the finite-dimensional shadow of the true infinite-dimensional difficulty.

Prioritized roadmap and limitations

The immediate path to a publishable Section 3I is now fairly clear.

The first milestone is to **freeze notation and prove only what is actually proved**. That means writing the exact denominator-aware detector $Z_{\alpha,\gamma}$, the seam-subtracted defect $D_{\alpha,\gamma}$, and the Separation Energy Lemma with explicit hypotheses on ω. This part is ready now.

The second milestone is to choose the host functional. The right choice is **Weil first**. Weil positivity is already RH-equivalent, already explicit-formula-native, and already quadratic. Bombieri's support-restricted variational work and later small-eigenvalue work around the Weil quadratic form strongly suggest that packet localization is not a fantasy but an analytically meaningful program.

The third milestone is to prove a **localized Gram theorem**: construct Mellin packets $g_{\alpha,\gamma}$ and show that the associated Weil Gram matrix is asymptotically diagonal on separated frequency windows. If that fails in a robust way, the failure itself is publishable, because it would identify a structural obstruction to the desired diagonal closure functional.

The fourth milestone is a fork:

- either prove a lower bound

$$Q_{\mathrm{Weil}}(g) \geq \sum_j c_j \, Q_{\omega_j}(\alpha_j, \gamma_j) - \varepsilon \parallel c \parallel^2,$$

  which yields the desired Q_{prime}, or
- prove a **no-go theorem** saying that no such diagonalization can hold in a broad natural class of packet systems.

Either outcome is mathematically valuable. The latter would stop the overreach cycle cleanly and redirect the program toward a genuinely different closure mechanism.

The backup program is Nyman–Beurling. If the Weil localization route proves too stiff, the right next move is to ask for a seam-projected distance in the Hardy/Mellin closure framework and to look for lower bounds controlled by Blaschke/inner factors attached to off-seam zeros. That is more global and perhaps less elegant from the detector viewpoint, but it is closer to an actual closure theorem by construction.

The principal risks are these.

The first risk is **cross-phase pollution**: the prime field is not a direct sum of independent zero families, and no classical theorem currently gives the packet-diagonal estimate needed for Section 3I.

The second risk is **denominator geometry**: the seam condition is not merely "$\cosh(\alpha u) - 1 > 0$" but the exact equality $1 - \rho = \bar{\rho}$. Any host functional that erases this distinction will likely be too coarse.

The third risk is **operator mismatch**: Gate B style two-fiber operators are aesthetically aligned with the functional equation, but unless they are tied rigorously to the explicit formula or a standard closure criterion, they remain models rather than proofs. The public Gate B note is an interesting blueprint, not yet a settled theorem source.

The fourth risk is **false victory by pointwise positivity**: the separation energy lemma is correct, but it is not itself RH. That local-to-global gap is the central limitation.

So the honest endpoint is:

The correct mathematical mold is now visible. The missing object is a global closure functional that diagonalizes or provably cannot d

That is where the work should concentrate next.